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LEARNER GUIDE

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Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 Jul 2026	First release. 36 hands-on labs mapped to CV0-004 domains 1–6.	Dr. Alfred Ang
2.0	19 Aug 2026	Major revision. Deck expanded with full per-domain teaching content covering every CV0-004 exam sub-objective (31 blueprint topics, 500+ visual slides); labs restructured into one folder per lab; free web-tool kit (IP Calculator, PCAP Analyzer, Regex Generator, Cybersecurity Simulator) integrated into the labs.	Dr. Alfred Ang

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How to Use This Guide

Welcome to the **WSQ CompTIA Certified Cloud+ Training (CV0-004)** hands-on Learner Guide, conducted by Tertiary Infotech Academy Pte Ltd (UEN 201200696W).

This guide contains **36 step-by-step labs** mapped to the six CompTIA Cloud+ CV0-004 exam domains. Every lab runs on the **free Killercoda Ubuntu Playground** — no local install, no cloud account and no credit card required.

Before you start:

- Open the Killercoda playground in your browser:
<https://killercoda.com/playgrounds/scenario/ubuntu>
- Work through the labs in order; each one builds on the mindset of the last.
- **Reset the playground** between labs that change firewall, container or kernel state.
- Copy the commands exactly as shown and read the short explanation under each step.
- The full lab source is also on GitHub:
<https://github.com/tertiarycourses/TGS-2024049214-CompTIA-Certified-Cloud-Training>

What You Will Achieve

- Explain cloud architecture, service models (IaaS/PaaS/SaaS/FaaS) and the shared responsibility model, and design highly-available, well-networked cloud solutions.
- Deploy cloud workloads using containers, virtualization, Infrastructure as Code and modern release strategies (blue-green, canary, rolling).
- Operate cloud systems with observability, logging, tracing, auto-scaling, backup/recovery and lifecycle management.
- Secure a cloud environment with vulnerability management, hardening, IAM/RBAC, secrets management, WAF and runtime threat detection.
- Apply DevOps fundamentals — source control, CI/CD, APIs and container orchestration.
- Troubleshoot common cloud deployment, network, security and container issues methodically.

Exam at a Glance

Item	Detail
Exam code	CV0-004
Questions	Maximum of 90
Question types	Multiple-choice and performance-based
Length	90 minutes
Passing score	750 (on a scale of 100–900)
Recommended experience	2–3 years as a systems administrator / cloud engineer; CompTIA Network+ and Server+ or equivalent knowledge

Domain	Exam Weight	Labs
1. Cloud Architecture	23%	1, 2, 3, 4, 5, 6, 7, 8

2. Deployment	19%	9, 10, 11, 12, 13, 14, 15
3. Operations	17%	16, 17, 18, 19, 20, 21
4. Security	19%	22, 23, 24, 25, 26, 27, 28
5. DevOps Fundamentals	10%	29, 30, 31, 32
6. Troubleshooting	12%	33, 34, 35, 36

Domain 1 — Cloud Architecture (23%)

Cloud concepts, service and deployment models, the shared responsibility model, high availability, networking (VPC), storage tiers, virtualization, databases and cloud-native / microservices design.

What this domain covers in the labs:

- **Lab 1 — Cloud Service Models & Shared Responsibility:** Run a local stand-in for each cloud service model and articulate who is responsible for what under the shared responsibility model.
- **Lab 2 — High Availability with HAProxy & Keepalived:** Build a health-checked load balancer across two AZ backends and add VRRP failover so a virtual IP survives a load-balancer failure.
- **Lab 3 — Cloud Networking with VPC Namespaces:** Construct a full VPC — two subnets, a router, a security-group rule and a NAT gateway — entirely from Linux network namespaces.
- **Lab 4 — Storage Tiers (Block, Object, File):** Create one block, one object and one file store locally, then benchmark IOPS to explain why storage tiers (and their prices) differ.
- **Lab 5 — Microservices & Service Discovery:** Deploy three loosely-coupled microservices behind Consul and demonstrate name-based service discovery, fan-out and resilience under a single failure.
- **Lab 6 — Containerization with Docker:** Practise every core containerization sub-objective — build/run/expose, ephemeral vs persistent storage, public/private registries and a Compose preview.
- **Lab 7 — Virtualization with QEMU/KVM:** Create and clone a QEMU VM to understand core virtualization concepts and contrast VMs with containers.
- **Lab 8 — Relational vs Non-Relational Databases:** Run and compare a self-managed relational DB against a document DB to learn when to choose each and how managed services shift responsibility.

Lab 1 — Cloud Service Models & Shared Responsibility

Goal: Run a local stand-in for each cloud service model and articulate who is responsible for what under the shared responsibility model.

Tools used: Docker, curl, jq, nginx, postgres:16, nextcloud

In this lab you will compare IaaS, PaaS, SaaS, and FaaS by running the equivalent of each model **locally** on the Killercode Ubuntu Playground. You will provision a raw VM-style workload (IaaS), deploy a managed Postgres-like service (PaaS), use a SaaS-style web app, and run a Function-as-a-Service handler. By the end you will be able to explain who is responsible for what under the **shared responsibility model**.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Update and install base tools

```
apt update && apt install -y docker.io curl jq python3-pip
systemctl start docker
```

Docker will let us simulate cloud-managed services as containers. **curl** and **jq** are used to call APIs.

Step 2 — IaaS simulation (you manage everything above the hypervisor)

Run a bare Ubuntu container — you patch it, you secure it, you install the runtime. This mirrors AWS EC2 / Azure VM / GCP Compute Engine.

```
docker run -dit --name iaas-vm ubuntu:22.04 bash
docker exec iaas-vm apt update
docker exec iaas-vm apt install -y nginx
docker exec iaas-vm service nginx start
```

You installed the OS package, configured the service, and started it. Under IaaS the **customer owns OS, runtime, app, data** and the **provider owns hardware, hypervisor, network**.

Step 3 — PaaS simulation (provider manages the runtime)

Run a managed-style Postgres. You only supply the database name and password — no OS patching.

```
docker run -d --name paas-db -e POSTGRES_PASSWORD=c1oud -p 5432:5432
postgres:16
sleep 5
docker exec paas-db psql -U postgres -c "CREATE DATABASE app;"
```

You did not install Postgres, configure **pg_hba.conf**, or apply OS patches. Under PaaS the **provider owns OS + runtime**, the **customer owns app + data**.

Step 4 — SaaS simulation (you only consume)

Run a fully packaged app — Nextcloud — the way you would consume Microsoft 365 or Salesforce.

```
docker run -d --name saas-app -p 8080:80 nextcloud:latest
sleep 10
curl -sI http://localhost:8080 | head -3
```

You did not write code, manage runtime, or touch the database. Under SaaS the **provider owns everything except your data and access**.

Step 5 — FaaS simulation (event-driven function)

Install OpenFaaS-style local handler with a Python script that runs on demand — like AWS Lambda.

```
mkdir -p /tmp/faas && cat > /tmp/faas/handler.py <<'EOF'
import sys, json
event = json.loads(sys.stdin.read())
print(json.dumps({"hello": event.get("name", "cloud")}))
EOF

echo '{"name": "Cloud+"}' | python3 /tmp/faas/handler.py
```

The function runs only when invoked, scales to zero, and bills per-invocation. The **provider owns the entire execution environment**; you only own the function code.

Step 6 — Map responsibility

Fill in the matrix from what you just ran:

Layer	IaaS	PaaS	SaaS	FaaS
Data	You	You	You	You
Application	You	You	Provider	You (code only)
Runtime	You	Provider	Provider	Provider
OS	You	Provider	Provider	Provider
Hypervisor / Hardware	Provider	Provider	Provider	Provider

Step 7 — Cleanup

```
docker rm -f iaas-vm paas-db saas-app
```

What you learned

- Hands-on difference between IaaS, PaaS, SaaS, and FaaS.
- Who patches the OS, who owns the data, and who runs the runtime in each model.
- The shared responsibility model is **never** "the provider does it all".

Free tools used

- Docker — <https://www.docker.com>
- KillerCoda Ubuntu Playground — <https://killercoda.com/playgrounds/scenario/ubuntu>
- AWS Shared Responsibility Model (reading) — <https://aws.amazon.com/compliance/shared-responsibility-model/>

Lab 2 — High Availability with HAProxy & Keepalived

Goal: Build a health-checked load balancer across two AZ backends and add VRRP failover so a virtual IP survives a load-balancer failure.

Tools used: HAProxy, Keepalived, Docker, nginx:alpine, curl

In this lab you will simulate a multi-AZ load-balanced deployment by running two backend web servers and an HAProxy load balancer in front of them. You will then add a second HAProxy instance with **Keepalived** so a virtual IP fails over when the primary

load balancer dies — the same pattern used by every cloud provider's regional load balancer.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install HAProxy and Keepalived

```
apt update && apt install -y haproxy keepalived docker.io curl
systemctl start docker
```

Step 2 — Launch two backend "AZ" web servers

Each container represents a server in a different Availability Zone.

```
docker run -d --name az1 -p 8081:80 nginx:alpine
docker run -d --name az2 -p 8082:80 nginx:alpine

docker exec az1 sh -c 'echo "AZ-1 healthy" > /usr/share/nginx/html/index.html'
docker exec az2 sh -c 'echo "AZ-2 healthy" > /usr/share/nginx/html/index.html'

curl http://localhost:8081
curl http://localhost:8082
```

Step 3 — Configure HAProxy with health checks

```
cat > /etc/haproxy/haproxy.cfg <<'EOF'
global
    daemon
    maxconn 256

defaults
    mode http
    timeout connect 5s
    timeout client 30s
    timeout server 30s

frontend web
    bind *:80
    default_backend azs

backend azs
    balance roundrobin
    option httpchk GET /
    server az1 127.0.0.1:8081 check
    server az2 127.0.0.1:8082 check
EOF

systemctl restart haproxy
```

Step 4 — Verify load balancing

```
for i in 1 2 3 4; do curl -s http://localhost/; done
```

You should see responses alternate between AZ-1 and AZ-2 — round-robin across availability zones.

Step 5 — Simulate an AZ failure

```
docker stop az1
sleep 3
for i in 1 2 3; do curl -s http://localhost/; done
```

HAProxy's health check marks **az1** down within seconds and routes 100% of traffic to AZ-2. This is the equivalent of an **AZ outage** in a cloud region.

```
docker start az1
```

Step 6 — Add Keepalived for VRRP failover

```
cat > /etc/keepalived/keepalived.conf <<'EOF'
vrrp_instance VI_1 {
    state MASTER
    interface eth0
    virtual_router_id 51
    priority 100
    advert_int 1
    authentication { auth_type PASS; auth_pass cloud }
    virtual_ipaddress { 10.0.0.250/24 }
}
EOF

systemctl start keepalived
ip -brief addr show eth0
```

The VIP **10.0.0.250** is now floating on this node. In a real two-node setup, when MASTER dies the BACKUP picks up the VIP within ~3 seconds — this is how regional **active-passive** load balancers achieve HA.

Step 7 — Cleanup

```
systemctl stop keepalived haproxy
docker rm -f az1 az2
```

What you learned

- How a cloud load balancer routes traffic across availability zones.
- How active health checks remove unhealthy instances automatically.
- How Keepalived/VRRP provides a floating IP for LB redundancy.

Free tools used

- HAProxy — <https://www.haproxy.org>
- Keepalived — <https://www.keepalived.org>
- Docker — <https://www.docker.com>

Lab 3 — Cloud Networking with VPC Namespaces

Goal: Construct a full VPC — two subnets, a router, a security-group rule and a NAT gateway — entirely from Linux network namespaces.

Tools used: iproute2 (ip netns), iptables, sysctl, ping

In this lab you will build a virtual private cloud (VPC) entirely with Linux network namespaces. You will create two subnets, a router, NAT to the internet, and a security group rule — exactly the building blocks AWS, Azure and GCP expose as managed services.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install tools

```
apt update && apt install -y iproute2 iptables iputils-ping
```

Step 2 — Create the VPC subnets

A network namespace is an isolated Linux network stack — equivalent to a subnet plus its attached ENIs.

```
ip netns add subnet-a
ip netns add subnet-b
ip netns add router

ip netns list
```

Step 3 — Wire the subnets to the router

Use veth pairs as virtual cables.

```
ip link add a-r type veth peer name r-a
ip link add b-r type veth peer name r-b

ip link set a-r netns subnet-a
ip link set r-a netns router
ip link set b-r netns subnet-b
ip link set r-b netns router
```

Assign IPs (10.0.1.0/24 and 10.0.2.0/24):

```
ip -n subnet-a addr add 10.0.1.10/24 dev a-r
ip -n router addr add 10.0.1.1/24 dev r-a
ip -n subnet-b addr add 10.0.2.10/24 dev b-r
ip -n router addr add 10.0.2.1/24 dev r-b

for ns in subnet-a subnet-b router; do
  ip -n $ns link set lo up
done

ip -n subnet-a link set a-r up
ip -n subnet-b link set b-r up
ip -n router link set r-a up
ip -n router link set r-b up
```

Step 4 — Add default routes (the route table)

```
ip -n subnet-a route add default via 10.0.1.1
ip -n subnet-b route add default via 10.0.2.1
ip netns exec router sysctl -w net.ipv4.ip_forward=1
```

Test inter-subnet connectivity (peering):

```
ip netns exec subnet-a ping -c 2 10.0.2.10
```

Step 5 — Add a security group rule (deny ICMP from B to A)

```
ip netns exec router iptables -I FORWARD -s 10.0.2.0/24 -d 10.0.1.0/24 -p icmp
-j DROP
ip netns exec subnet-b ping -c 2 -W 2 10.0.1.10 || echo "Blocked by SG"
```

This is the equivalent of an AWS Security Group / Azure NSG / GCP Firewall rule.

Step 6 — Add NAT gateway to the internet

```
ip link add nat-r type veth peer name r-nat
ip link set r-nat netns router
ip addr add 192.168.99.1/24 dev nat-r
ip -n router addr add 192.168.99.2/24 dev r-nat
ip link set nat-r up
ip -n router link set r-nat up
ip -n router route add default via 192.168.99.1

sysctl -w net.ipv4.ip_forward=1
iptables -t nat -A POSTROUTING -s 10.0.0.0/16 -o eth0 -j MASQUERADE

ip netns exec subnet-a ping -c 2 8.8.8.8 || echo "Outbound test"
```

This mirrors a cloud **NAT Gateway** for private subnets reaching the internet.

Step 7 — Cleanup

```
ip netns del subnet-a
ip netns del subnet-b
ip netns del router
ip link del nat-r 2>/dev/null
iptables -t nat -F POSTROUTING
```

What you learned

- VPCs, subnets, route tables, security groups, and NAT gateways are kernel features.
- Peering = a route + a forwarding rule.
- Security groups operate per-flow at L3/L4.

Free tools used

- iproute2 / iptables (built-in)
- TIS IP Calculator — <https://alfredang.github.io/ipcalculator/>

Lab 4 — Storage Tiers (Block, Object, File)

Goal: Create one block, one object and one file store locally, then benchmark IOPS to explain why storage tiers (and their prices) differ.

Tools used: fio, losetup/mkfs.ext4, MinIO + mc, NFS, Docker

In this lab you will create one of each cloud storage type on the Killercoda VM: a **block** device with a loopback file, an **object** store with MinIO (S3-compatible), and a **file** share with NFS. You will then measure IOPS with **fio** to see why hot/warm/cold tiering exists.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install tools

```
apt update && apt install -y fio nfs-kernel-server nfs-common docker.io
systemctl start docker
```

Step 2 — Create a BLOCK device (like AWS EBS)

```
dd if=/dev/zero of=/tmp/disk.img bs=1M count=256
LOOP=$(losetup -f --show /tmp/disk.img)
mkfs.ext4 $LOOP
mkdir -p /mnt/block
mount $LOOP /mnt/block
df -h /mnt/block
```

Block storage = raw device, formatted with a filesystem, mounted to a single VM.

Step 3 — Create an OBJECT store (like AWS S3)

```
docker run -d --name minio -p 9000:9000 -p 9001:9001 \
-e MINIO_ROOT_USER=admin -e MINIO_ROOT_PASSWORD=cloudplus \
minio/minio server /data --console-address ":9001"

sleep 5

docker exec minio mc alias set local http://127.0.0.1:9000 admin cloudplus
docker exec minio mc mb local/hot-bucket
echo "Hello Cloud+" > /tmp/file.txt
docker cp /tmp/file.txt minio:/file.txt
docker exec minio mc cp /file.txt local/hot-bucket/
docker exec minio mc ls local/hot-bucket/
```

Object storage = HTTP API, infinite scale, no filesystem, accessed by key.

Step 4 — Create a FILE share (like AWS EFS / Azure Files)

```
mkdir -p /srv/nfs && chmod 777 /srv/nfs
echo '/srv/nfs *(rw, sync, no_subtree_check, no_root_squash)' > /etc/exports
exportfs -ra
systemctl start nfs-kernel-server

mkdir -p /mnt/file
mount -t nfs 127.0.0.1:/srv/nfs /mnt/file
echo "shared" > /mnt/file/test.txt
cat /srv/nfs/test.txt
```

File storage = POSIX filesystem, shared by many clients over NFS or SMB.

Step 5 — Compare IOPS between tiers (hot vs cold)

```
echo "--- BLOCK (hot, SSD-like) ---"
fio --name=hot --filename=/mnt/block/test --size=64M --rw=randwrite --bs=4k --
runtime=5 --time_based --group_reporting | grep IOPS

echo "--- FILE over NFS (warm) ---"
fio --name=warm --filename=/mnt/file/test --size=64M --rw=randwrite --bs=4k --
runtime=5 --time_based --group_reporting | grep IOPS
```

You will see block storage delivers significantly higher random-write IOPS than NFS — this is why **hot** tier (SSD block) costs more than **warm** (file) and far more than **cold/archive** (object).

Step 6 — Tier mapping

Tier	Cloud example	This lab
Hot (SSD block)	AWS gp3, Azure Premium SSD	<code>/mnt/block</code> (ext4)
Warm (file)	AWS EFS, Azure Files	<code>/mnt/file</code> (NFS)
Cold (object std)	AWS S3 Standard	MinIO <code>hot-bucket</code>
Archive	AWS Glacier, Azure Archive	MinIO with lifecycle

Step 7 — Cleanup

```
umount /mnt/block /mnt/file
losetup -d $LOOP
docker rm -f minio
systemctl stop nfs-kernel-server
```

What you learned

- Block, object, and file storage have different APIs and access patterns.
- IOPS varies by an order of magnitude between tiers.
- Cost models follow performance — hot is fastest and most expensive.

Free tools used

- MinIO — <https://min.io>
- NFS kernel server (built-in)
- fio — <https://github.com/axboe/fio>

Lab 5 — Microservices & Service Discovery

Goal: Deploy three loosely-coupled microservices behind Consul and demonstrate name-based service discovery, fan-out and resilience under a single failure.

Tools used: HashiCorp Consul, Docker bridge network, curl, jq, dig

In this lab you will deploy three loosely-coupled microservices behind Consul for service discovery. You will see how containers register themselves, how a client resolves a service by name (not IP), and how the fan-out pattern works when one service calls many.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker and create a network

```
apt update && apt install -y docker.io curl jq
systemctl start docker
docker network create cloudnet
```

Step 2 — Run Consul (service registry)

```
docker run -d --name consul --network cloudnet \
  -p 8500:8500 hashicorp/consul:latest \
  agent -dev -client=0.0.0.0
sleep 3
curl -s http://localhost:8500/v1/status/leader
```

Step 3 — Deploy three microservices

Each service is a tiny HTTP responder.

```
for n in users orders payments; do
  docker run -d --name svc-$n --network cloudnet \
    -e PORT=80 \
    nginx:alpine
  docker exec svc-$n sh -c "echo '{\"service\": \"$n\", \"ok\": true}' >
  /usr/share/nginx/html/index.html"
done
```

Step 4 — Register them with Consul

```
for n in users orders payments; do
  curl -s -X PUT -d "{
    \"Name\": \"$n\",
    \"Address\": \"svc-$n\",
    \"Port\": 80,
    \"Check\": {\"HTTP\": \"http://svc-$n/\", \"Interval\": \"10s\"}
  }" http://localhost:8500/v1/agent/service/register
done

curl -s http://localhost:8500/v1/catalog/services | jq
```

Step 5 — Discover a service by name (DNS interface)

Consul exposes service names over DNS on port 8600.

```
docker run --rm --network cloudnet --dns 172.17.0.1 alpine \
  sh -c "apk add --quiet bind-tools && dig @consul -p 8600
  users.service.consul"
```

Clients now resolve `users.service.consul` instead of hard-coded IPs — the **service discovery** primitive of every cloud-native platform.

Step 6 — Fan-out call

Simulate one request triggering parallel calls to all three services (event fan-out).

```
docker run --rm --network cloudnet alpine sh -c "
  apk add --quiet curl
  for s in users orders payments; do
    (curl -s http://svc-$s/ &)
  done
  wait
"
```

Step 7 — Loosely coupled: kill one service, others survive

```
docker stop svc-orders
curl -s http://localhost:8500/v1/health/service/users | jq '.
[0].Checks[0].Status'
curl -s http://localhost:8500/v1/health/service/orders | jq '.
[0].Checks[0].Status'
```

users stays **passing**, **orders** flips to **critical** — but the rest of the system keeps running. That is **loose coupling**.

Step 8 — Cleanup

```
docker rm -f consul svc-users svc-orders svc-payments
docker network rm cloudnet
```

What you learned

- Microservices register and deregister with a service registry.
- Clients resolve services by name, not IP — IPs come and go.
- Fan-out and loose coupling are key cloud-native design patterns.

Free tools used

- HashiCorp Consul — <https://www.consul.io>
- Docker — <https://www.docker.com>

Lab 6 — Containerization with Docker

Goal: Practise every core containerization sub-objective — build/run/expose, ephemeral vs persistent storage, public/private registries and a Compose preview.

Tools used: Docker, Dockerfile, registry:2, Docker Compose

In this lab you will build a custom image, run a stand-alone container, expose ports, mount persistent and ephemeral storage, and push to a local registry. By the end you will have practised every CV0-004 containerization sub-objective.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker

```
apt update && apt install -y docker.io
systemctl start docker
docker --version
```


Step 2 — Build a custom image

```
mkdir -p /tmp/app && cd /tmp/app
cat > index.html <<'EOF'
<h1>Cloud+ container</h1>
EOF
cat > Dockerfile <<'EOF'
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
EOF

docker build -t myapp:1.0 .
docker images | grep myapp
```

Step 3 — Stand-alone run with port mapping

```
docker run -d --name web -p 8080:80 myapp:1.0
curl -s http://localhost:8080
```

-p 8080:80 maps host port 8080 → container port 80. This is **port mapping**, a CV0-004 sub-objective.

Step 4 — Ephemeral storage (default)

```
docker exec web sh -c 'echo lost > /tmp/scratch.txt'
docker exec web cat /tmp/scratch.txt
docker rm -f web
docker run -d --name web -p 8080:80 myapp:1.0
docker exec web ls /tmp/scratch.txt 2>&1 || echo "Gone – ephemeral"
```

The file disappeared with the container — that is **ephemeral storage**.

Step 5 — Persistent volume

```
docker volume create app-data
docker rm -f web
docker run -d --name web -p 8080:80 \
  -v app-data:/data myapp:1.0
docker exec web sh -c 'echo durable > /data/keep.txt'
docker rm -f web
docker run --rm -v app-data:/data alpine cat /data/keep.txt
```

The file survived a container delete — that is a **persistent volume**.

Step 6 — Pull from a public image registry

```
docker pull alpine:3.20
docker pull hello-world
docker images
```

[docker.io](https://www.docker.com) is a **public** registry. You can also push to private registries like ECR, ACR, GAR.

Step 7 — Run a private registry

```
docker run -d --name registry -p 5000:5000 registry:2
docker tag myapp:1.0 localhost:5000/myapp:1.0
```

```
docker push localhost:5000/myapp:1.0
docker pull localhost:5000/myapp:1.0
curl -s http://localhost:5000/v2/_catalog
```

Step 8 — Workload orchestration preview (Compose)

```
apt install -y docker-compose-v2 || apt install -y docker-compose
cat > /tmp/docker-compose.yml <<'EOF'
services:
  web:
    image: myapp:1.0
    ports: ["8081:80"]
  cache:
    image: redis:7-alpine
EOF
cd /tmp && docker compose up -d
docker compose ps
```

This previews **workload orchestration** — covered fully in Lab 32 (Kubernetes).

Step 9 — Cleanup

```
cd / && docker compose -f /tmp/docker-compose.yml down
docker rm -f web registry
docker volume rm app-data
```

What you learned

- Build, tag, run, and push images.
- Port mapping vs persistent volumes vs ephemeral storage.
- Public vs private image registries.

Free tools used

- Docker Engine — <https://www.docker.com>
- Docker Hub — <https://hub.docker.com>
- Distribution registry — <https://github.com/distribution/distribution>

Lab 7 — Virtualization with QEMU/KVM

Goal: Create and clone a QEMU VM to understand core virtualization concepts and contrast VMs with containers.

Tools used: QEMU, qemu-img, libvirt, bridge-utils, Alpine ISO

In this lab you will create a virtual machine with QEMU, examine virtualization concepts (clone, host affinity, hardware pass-through, VM networks), and compare it to the container model from Lab 6.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

*Note: Killercoda runs nested under KVM, so we use **QEMU TCG** (software emulation) instead of full KVM acceleration. The concepts and commands are identical to a real hypervisor.*

Step 1 — Install QEMU and libvirt tools

```
apt update && apt install -y qemu-system-x86 qemu-utils libvirt-clients  
libvirt-daemon-system bridge-utils virtinst cloud-image-utils
```

Step 2 — Create a virtual disk (VM storage, local)

```
mkdir -p /var/vm && cd /var/vm  
qemu-img create -f qcow2 vm1.qcow2 2G  
qemu-img info vm1.qcow2
```

qcow2 is a copy-on-write disk format — the basis of fast **cloning**.

Step 3 — Clone the disk (linked clone)

```
qemu-img create -f qcow2 -F qcow2 -b vm1.qcow2 vm2.qcow2  
ls -lh vm*.qcow2  
qemu-img info vm2.qcow2
```

vm2.qcow2 is a thin clone — only stores deltas. This is how cloud snapshots and templates work.

Step 4 — Examine the host's virtualization capability

```
egrep -c '(vmx|svm)' /proc/cpuinfo  
lscpu | grep -i virtual
```

If **vmx** (Intel) or **svm** (AMD) is non-zero, the host supports hardware-assisted virtualization (KVM). That is **hardware pass-through** at the CPU level.

Step 5 — Boot a tiny VM

```
cd /var/vm  
wget -q https://dl-cdn.alpinelinux.org/alpine/v3.19/releases/x86_64/alpine-  
virt-3.19.1-x86_64.iso  
  
timeout 30 qemu-system-x86_64 \  
-m 256 \  
-nographic \  
-hda vm1.qcow2 \  
-cdrom alpine-virt-3.19.1-x86_64.iso \  
-boot d \  
-net nic -net user 2>&1 | head -30 || true
```

The VM boots its own kernel — unlike a container, which shares the host kernel.

Step 6 — VM network types

QEMU shows the two main virtual network types from the exam:

```
echo "--- User-mode (NAT, default) ---"  
echo '-net user creates an isolated NAT – like an overlay network'  
  
echo "--- Bridged (VM joins host LAN) ---"  
brctl addbr br0 2>/dev/null || ip link add br0 type bridge  
ip link show br0  
ip link del br0
```

- **Overlay** = `-net user` — VM traffic encapsulated, NATed.
- **VM network (bridged)** = bridge — VMs appear on the same L2 segment as the host.

Step 7 — Host affinity

In libvirt you pin a VM to a specific host CPU set:

```
echo '<vcpu placement="static" cpuset="0-1">2</vcpu>'
```

`cpuset` = host affinity. The same concept on AWS is **Dedicated Host** + placement groups.

Step 8 — VM vs Container summary

Aspect	VM (this lab)	Container (Lab 6)
Kernel	Own kernel	Shares host kernel
Boot time	seconds–minutes	milliseconds
Disk	qcow2 / raw	layered images
Isolation	Hardware-level	Namespace + cgroup
Density per host	Tens	Thousands

Step 9 — Cleanup

```
rm -f /var/vm/vm*.qcow2 /var/vm/alpine-virt-3.19.1-x86_64.iso
```

What you learned

- Create and clone qcow2 disks.
- VM networks: user-mode, bridged, overlay.
- VMs vs containers — when to choose which.

Free tools used

- QEMU — <https://www.qemu.org>
- libvirt / virt-install — <https://libvirt.org>
- Alpine Linux ISO — <https://alpinelinux.org>

Lab 8 — Relational vs Non-Relational Databases

Goal: Run and compare a self-managed relational DB against a document DB to learn when to choose each and how managed services shift responsibility.

Tools used: Docker, PostgreSQL 16 + psql, MongoDB + mongosh

In this lab you will run a **self-managed** Postgres (relational) and a **provider-managed-style** MongoDB (non-relational), compare schemas and queries, and discuss when to choose each.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker

```
apt update && apt install -y docker.io postgresql-client  
systemctl start docker
```

Step 2 — Self-managed Postgres (you handle everything)

```
docker run -d --name pg \  
-e POSTGRES_PASSWORD=cloud \  
-p 5432:5432 \  
postgres:16  
  
sleep 5  
PGPASSWORD=cloud psql -h 127.0.0.1 -U postgres -c "SELECT version();"
```

You picked the version, set the password, manage backups. **Self-managed** = customer-owned lifecycle.

Step 3 — Create a relational schema

```
PGPASSWORD=cloud psql -h 127.0.0.1 -U postgres <<'SQL'  
CREATE TABLE customers (  
  id SERIAL PRIMARY KEY,  
  email TEXT UNIQUE NOT NULL,  
  name TEXT  
);  
CREATE TABLE orders (  
  id SERIAL PRIMARY KEY,  
  customer_id INT REFERENCES customers(id),  
  amount NUMERIC(10,2)  
);  
INSERT INTO customers(email, name) VALUES ('a@x.com','Alice'),  
('b@x.com','Bob');  
INSERT INTO orders(customer_id, amount) VALUES (1, 12.50), (1, 99.00), (2,  
5.00);  
  
SELECT c.name, SUM(o.amount) AS total  
FROM customers c JOIN orders o ON o.customer_id = c.id  
GROUP BY c.name;  
SQL
```

Strict schema, foreign keys, JOIN — relational strengths.

Step 4 — Provider-managed-style MongoDB

```
docker run -d --name mongo -p 27017:27017 mongo:7  
sleep 5  
docker exec -i mongo mongosh <<'JS'  
use shop  
db.customers.insertOne({email:"a@x.com", name:"Alice", orders:[  
  {amount: 12.50, item:"pen"},  
  {amount: 99.00, item:"chair"}  
]})  
db.customers.insertOne({email:"b@x.com", name:"Bob", orders:[  
  {amount: 5.00, item:"pad"}  
]})
```

```

db.customers.aggregate([
  {$unwind:"$orders"},
  {$group: {_id:"$name", total:{$sum:"$orders.amount"}}}
])
JS

```

No schema enforced, nested documents, no JOIN. Suitable for variable-shape data.

Step 5 — Compare strengths

Need	Relational (Postgres)	Non-relational (Mongo)
Strict schema, ACID	✓	partial
Multi-table JOIN	✓	✗
Variable structure	✗	✓
Horizontal scale-out	harder	built-in sharding
Examples	Banking, ERP	IoT, catalog, sessions

Step 6 — Provider-managed equivalents

Engine	AWS	Azure	GCP
Postgres	RDS / Aurora	Azure Database for Postgres	Cloud SQL
MongoDB	DocumentDB	Cosmos DB (Mongo API)	Firestore
Redis	ElastiCache	Azure Cache for Redis	Memorystore

In **provider-managed** the cloud handles patching, HA, backups; you only manage data and connections.

Step 7 — Cleanup

```
docker rm -f pg mongo
```

What you learned

- Relational vs non-relational data models.
- Self-managed vs provider-managed responsibility split.
- When to pick each engine.

Free tools used

- PostgreSQL — <https://www.postgresql.org>
- MongoDB Community — <https://www.mongodb.com/try/download/community>
- DB Fiddle (web SQL sandbox) — <https://www.db-fiddle.com>

Domain 2 — Deployment (19%)

Provisioning and configuring cloud resources: deployment models, zero-downtime release strategies (blue-green, canary, rolling), Infrastructure as Code, Configuration as Code and scripting logic.

What this domain covers in the labs:

- **Lab 9 — Public, Private & Hybrid Cloud Models:** Simulate public (LocalStack), private (MinIO) and hybrid deployment models to compare their cost, control and elasticity trade-offs.
- **Lab 10 — Blue-Green Deployment with Nginx:** Run blue (v1) and green (v2) side by side and switch live traffic instantly via an Nginx reload to achieve zero-downtime deployment.
- **Lab 11 — Canary Deployment with HAProxy:** Gradually shift live traffic from v1 to v2 using HAProxy server weights to safely validate a new release before full promotion.
- **Lab 12 — Rolling Deployment with Docker Compose:** Deploy three load-balanced replicas and upgrade them one at a time to perform a zero-downtime rolling update.
- **Lab 13 — Infrastructure as Code with Terraform:** Write a Terraform config that provisions an S3 bucket on LocalStack and experience plan/apply, state, drift detection and versioning.
- **Lab 14 — Configuration as Code with Ansible:** Write an Ansible playbook that installs and configures Nginx idempotently and prove drift correction.
- **Lab 15 — JSON & YAML Scripting Logic:** Parse, transform, validate and convert JSON ↔ YAML while exercising the exam's scripting-logic primitives.

Lab 9 — Public, Private & Hybrid Cloud Models

Goal: Simulate public (LocalStack), private (MinIO) and hybrid deployment models to compare their cost, control and elasticity trade-offs.

Tools used: LocalStack, MinIO + mc, AWS CLI, WireGuard, Docker

In this lab you will simulate a **private** cloud (your Killercode VM), a **public** cloud (LocalStack — a free local AWS), and connect them as a **hybrid** with a WireGuard tunnel. By the end you will see why each model trades cost, control, and elasticity differently.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install tools

```
apt update && apt install -y docker.io awscli wireguard wireguard-tools
systemctl start docker
```

Step 2 — Stand up a "public" cloud with LocalStack

LocalStack is a free, open-source AWS emulator — perfect for the lab without a real account.

```
docker run -d --name public-cloud -p 4566:4566 \
  -e SERVICES=s3,ec2,iam \
  localstack/localstack:latest
sleep 8
curl -s http://localhost:4566/_localstack/health | head -c 200
```

Configure the AWS CLI to talk to it:

```
aws configure set aws_access_key_id test
aws configure set aws_secret_access_key test
aws configure set default.region us-east-1

aws --endpoint-url=http://localhost:4566 s3 mb s3://public-bucket
aws --endpoint-url=http://localhost:4566 s3 ls
```

This is your **public cloud** — multi-tenant API, region-based, pay-per-use.

Step 3 — Stand up a "private" cloud (on-prem)

Use a local MinIO bucket — the same S3 API, but inside your data centre.

```
docker run -d --name private-cloud -p 9000:9000 \
  -e MINIO_ROOT_USER=admin -e MINIO_ROOT_PASSWORD=cloudplus \
  minio/minio server /data

sleep 4
docker exec private-cloud mc alias set local http://127.0.0.1:9000 admin
cloudplus
docker exec private-cloud mc mb local/private-bucket
```

This is **private** / on-prem — single tenant, full control.

Step 4 — Hybrid: copy data between them

```
echo "hybrid-payload" > /tmp/data.txt
aws --endpoint-url=http://localhost:4566 s3 cp /tmp/data.txt s3://public-
bucket/
docker cp /tmp/data.txt private-cloud:/data.txt
docker exec private-cloud mc cp /data.txt local/private-bucket/

aws --endpoint-url=http://localhost:4566 s3 ls s3://public-bucket/
docker exec private-cloud mc ls local/private-bucket/
```

The same object now lives in both clouds — the basis of a **hybrid** model.

Step 5 — Community cloud concept

A community cloud is shared by a group with common compliance needs (e.g. healthcare, government). You would model it as a private cloud with **multi-tenant access** restricted to the community. No infra to deploy — just policy.

Step 6 — Decision matrix

Factor	Public	Private	Hybrid	Community
CapEx	Low	High	Mixed	Shared
Elasticity	Best	Limited	Burst-able	Limited
Control	Lowest	Highest	Mixed	Shared
Compliance	varies	strongest	depends	strongest for sector

Step 7 — Cleanup

```
docker rm -f public-cloud private-cloud
```


What you learned

- Public, private, hybrid, and community models differ in tenancy and control.
- The **API** can stay the same (S3) across deployment models.
- Hybrid lets you burst from on-prem to public cloud.

Free tools used

- LocalStack Community — <https://www.localstack.cloud>
- MinIO — <https://min.io>
- AWS CLI — <https://aws.amazon.com/cli>

Lab 10 — Blue-Green Deployment with Nginx

Goal: Run blue (v1) and green (v2) side by side and switch live traffic instantly via an Nginx reload to achieve zero-downtime deployment.

Tools used: Nginx, Docker (nginx:alpine), curl

In this lab you will run two versions of an application — the live "blue" v1 and the staged "green" v2 — and switch traffic instantly by reloading Nginx. This is the canonical zero-downtime deployment pattern.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Nginx and Docker

```
apt update && apt install -y nginx docker.io curl
systemctl start docker
systemctl start nginx
```

Step 2 — Run Blue (v1) and Green (v2)

```
docker run -d --name blue -p 8081:80 nginx:alpine
docker run -d --name green -p 8082:80 nginx:alpine

docker exec blue sh -c 'echo "v1 BLUE" > /usr/share/nginx/html/index.html'
docker exec green sh -c 'echo "v2 GREEN" > /usr/share/nginx/html/index.html'
```

Step 3 — Point Nginx at Blue

```
cat > /etc/nginx/sites-available/default <<'EOF'
upstream live { server 127.0.0.1:8081; } # BLUE
server {
    listen 80;
    location / { proxy_pass http://live; }
}
EOF
nginx -t && systemctl reload nginx
curl -s http://localhost/
```

Step 4 — Smoke-test Green out of band

```
curl -s http://localhost:8082/
```

Green is fully running but receives **zero customer traffic**. Run integration tests here.

Step 5 — Switch traffic to Green (cutover)

```
sed -i 's/8081/8082/' /etc/nginx/sites-available/default
nginx -t && systemctl reload nginx
for i in 1 2 3; do curl -s http://localhost/; done
```

nginx -s reload swaps without dropping a connection. Cutover is **atomic**.

Step 6 — Instant rollback

If v2 is broken, switch back in one command:

```
sed -i 's/8082/8081/' /etc/nginx/sites-available/default
nginx -t && systemctl reload nginx
curl -s http://localhost/
```

This is why blue-green is preferred for high-risk releases.

Step 7 — Cleanup

```
docker rm -f blue green
systemctl stop nginx
```

What you learned

- Blue-green keeps two parallel environments; cutover is a config change.
- Rollback is symmetrical and fast.
- Smoke-test the new version on its own port before flipping traffic.

Free tools used

- Nginx — <https://nginx.org>
- Docker — <https://www.docker.com>

Lab 11 — Canary Deployment with HAProxy

Goal: Gradually shift live traffic from v1 to v2 using HAProxy server weights to safely validate a new release before full promotion.

Tools used: HAProxy, Docker (nginx:alpine), curl

In this lab you will gradually shift traffic from v1 to v2 using HAProxy server weights — 90/10, then 50/50, then 100% — exactly the strategy used by AWS ALB weighted target groups, Istio canaries, and Argo Rollouts.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install HAProxy and Docker

```
apt update && apt install -y haproxy docker.io curl
systemctl start docker
```

Step 2 — Run v1 (stable) and v2 (canary)

```
docker run -d --name v1 -p 8091:80 nginx:alpine
docker run -d --name v2 -p 8092:80 nginx:alpine
docker exec v1 sh -c 'echo v1 > /usr/share/nginx/html/index.html'
docker exec v2 sh -c 'echo v2 > /usr/share/nginx/html/index.html'
```

Step 3 — HAProxy at 90/10 (canary 10%)

```
cat > /etc/haproxy/haproxy.cfg <<'EOF'
global
    daemon
defaults
    mode http
    timeout connect 5s
    timeout client 30s
    timeout server 30s
frontend fe
    bind *:80
    default_backend app
backend app
    balance roundrobin
    server v1 127.0.0.1:8091 weight 90 check
    server v2 127.0.0.1:8092 weight 10 check
EOF
systemctl restart haproxy
```

Verify the split:

```
for i in $(seq 1 50); do curl -s http://localhost/; done | sort | uniq -c
```

You should see ~45 v1 / ~5 v2.

Step 4 — Increase to 50/50

```
sed -i 's/weight 90/weight 50/; s/weight 10/weight 50/'
/etc/haproxy/haproxy.cfg
systemctl reload haproxy
```

```
for i in $(seq 1 50); do curl -s http://localhost/; done | sort | uniq -c
```

Step 5 — Promote v2 to 100%

```
sed -i 's/weight 50/weight 0/; 0,/weight 0/!s/weight 0/weight 100/'
/etc/haproxy/haproxy.cfg
# Simpler: rewrite cleanly
cat > /etc/haproxy/haproxy.cfg <<'EOF'
global
    daemon
defaults
    mode http
    timeout connect 5s
    timeout client 30s
    timeout server 30s
frontend fe
    bind *:80
    default_backend app
```

```

backend app
    server v1 127.0.0.1:8091 weight 0 check backup
    server v2 127.0.0.1:8092 weight 100 check
EOF
systemctl reload haproxy

for i in $(seq 1 20); do curl -s http://localhost/; done | sort | uniq -c

```

100% v2. Roll-out complete.

Step 6 — Rollback (kill the canary)

If your monitoring (Lab 16) shows v2 errors rising, flip the weights back:

```

sed -i 's/weight 0 check backup/weight 100 check/; s/weight 100 check$/weight
0 check backup/' /etc/haproxy/haproxy.cfg
systemctl reload haproxy
curl -s http://localhost/

```

Step 7 — Cleanup

```

docker rm -f v1 v2
systemctl stop haproxy

```

What you learned

- Canary = small percentage of real traffic on a new version.
- Weight-based routing lets you ramp gradually and rollback instantly.
- Always pair canary with metrics/alerts.

Free tools used

- HAProxy — <https://www.haproxy.org>
- Docker — <https://www.docker.com>

Lab 12 — Rolling Deployment with Docker Compose

Goal: Deploy three load-balanced replicas and upgrade them one at a time to perform a zero-downtime rolling update.

Tools used: Docker Compose, HAProxy, curl

In this lab you will deploy three replicas of an app behind a load balancer, then upgrade them **one at a time** while traffic continues — the rolling deployment strategy used by Kubernetes Deployments, ECS services, and ASGs with rolling updates.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker Compose

```

apt update && apt install -y docker.io docker-compose-v2 curl
systemctl start docker

```

Step 2 — Define the stack

```
mkdir -p /tmp/rolling && cd /tmp/rolling
cat > docker-compose.yml <<'EOF'
services:
  app:
    image: nginx:alpine
    deploy:
      replicas: 3
      networks: [back]
    command: >
      sh -c 'echo "v1 from $$HOSTNAME" > /usr/share/nginx/html/index.html &&
nginx -g "daemon off;"'
  lb:
    image: haproxy:alpine
    ports: ["80:80"]
    volumes: ["/haproxy.cfg:/usr/local/etc/haproxy/haproxy.cfg:ro"]
    networks: [back]
networks:
  back:
EOF

cat > haproxy.cfg <<'EOF'
defaults
  mode http
  timeout connect 3s
  timeout client 30s
  timeout server 30s
frontend fe
  bind *:80
  default_backend app
backend app
  balance roundrobin
  server-template app- 3 app:80 check resolvers docker init-addr libc,none
resolvers docker
  nameserver dns 127.0.0.11:53
EOF

docker compose up -d --scale app=3
sleep 4
for i in 1 2 3 4 5 6; do curl -s http://localhost/; done
```

Step 3 — Upgrade replicas one at a time

```
# Replica 1
CID=$(docker compose ps -q app | head -n1)
docker stop $CID && docker rm $CID
docker compose up -d --scale app=3 --no-recreate
sleep 2

# Replica 2
CID=$(docker compose ps -q app | sed -n 2p)
docker stop $CID && docker rm $CID
docker compose up -d --scale app=3 --no-recreate
```

```
sleep 2

# Replica 3
CID=$(docker compose ps -q app | tail -n1)
docker stop $CID && docker rm $CID
docker compose up -d --scale app=3 --no-recreate
sleep 2
```

While each is replaced, the other two keep serving — no downtime.

```
for i in $(seq 1 9); do curl -s http://localhost/; done
```

Step 4 — Compare strategies

Strategy	Resource use	Risk window	Rollback
In-place	1×	Total downtime during swap	Restore + redeploy
Rolling	1×–1.x×	Mixed versions during roll	Roll backwards
Blue-Green	2×	Zero (atomic flip)	Instant
Canary	1×–1.x×	Small % users	Reset weights

Step 5 — Cleanup

```
cd /tmp/rolling && docker compose down
```

What you learned

- Rolling = replace replicas one (or batch) at a time.
- Cheaper than blue-green but produces a brief mixed-version window.
- Always pair with a readiness check before progressing.

Free tools used

- Docker Compose — <https://docs.docker.com/compose>
- HAProxy — <https://www.haproxy.org>

Lab 13 — Infrastructure as Code with Terraform

Goal: Write a Terraform config that provisions an S3 bucket on LocalStack and experience plan/apply, state, drift detection and versioning.

Tools used: Terraform, LocalStack, awslocal, Git, Docker

In this lab you will write a Terraform configuration that provisions an S3 bucket and a security policy on **LocalStack** (free local AWS). You will see plan/apply, state, drift detection, and versioning.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Terraform and LocalStack

```
apt update && apt install -y wget unzip docker.io
systemctl start docker
```

```
wget -q
https://releases.hashicorp.com/terraform/1.9.5/terraform_1.9.5_linux_amd64.zip
unzip -o terraform_1.9.5_linux_amd64.zip -d /usr/local/bin
terraform version

docker run -d --name localstack -p 4566:4566 -e SERVICES=s3,iam
localstack/localstack:latest
sleep 8
```

Step 2 — Write the Terraform configuration

```
mkdir -p /tmp/tf && cd /tmp/tf
cat > main.tf <<'EOF'
terraform {
  required_providers { aws = { source = "hashicorp/aws" version = "~> 5.0" } }
}

provider "aws" {
  region                = "us-east-1"
  access_key            = "test"
  secret_key            = "test"
  skip_credentials_validation = true
  skip_metadata_api_check  = true
  skip_requesting_account_id = true
  s3_use_path_style        = true
  endpoints { s3 = "http://localhost:4566" iam = "http://localhost:4566" }
}

variable "env" { default = "dev" }

resource "aws_s3_bucket" "data" {
  bucket = "cloudplus-${var.env}-data"
}
EOF
```

Step 3 — Init, plan, apply

```
terraform init
terraform plan
terraform apply -auto-approve
```

Inspect state:

```
terraform state list
terraform state show aws_s3_bucket.data
ls -lh terraform.tfstate
```

The state file is the source of truth — back it up to a remote backend in production.

Step 4 — Drift detection

Manually mutate the bucket outside Terraform:

```
docker exec localstack awslocal s3api put-bucket-tagging \
  --bucket cloudplus-dev-data \
  --tagging 'TagSet=[{Key=DriftedBy,Value=human}]'
```

```
terraform plan
```

terraform plan reports the **drift** — Terraform wants to remove the manual tag.

Step 5 — Version your code (Git)

```
apt install -y git
cd /tmp/tf
git init -q
echo 'terraform.tfstate*' > .gitignore
echo '.terraform/' >> .gitignore
git add . && git -c user.email=lab@x -c user.name=lab commit -q -m "initial
infra"
git log --oneline
```

Code reviewable, diff-able, rollback-able.

Step 6 — Apply a change (test → apply pattern)

```
sed -i 's/env" { default = "dev"/env" { default = "prod"/' main.tf
terraform plan
terraform apply -auto-approve
terraform state list
```

Terraform creates **cloudplus-prod-data**. The dev one is destroyed because the resource name is parameterised — typical IaC behaviour.

Step 7 — Cleanup

```
terraform destroy -auto-approve
docker rm -f localstack
```

What you learned

- IaC: declarative config → cloud resources.
- Plan before apply.
- Drift detection catches manual changes.
- State must be stored safely.

Free tools used

- Terraform — <https://developer.hashicorp.com/terraform>
- LocalStack Community — <https://www.localstack.cloud>
- OpenTofu (Terraform fork) — <https://opentofu.org>

Lab 14 — Configuration as Code with Ansible

Goal: Write an Ansible playbook that installs and configures Nginx idempotently and prove drift correction.

Tools used: Ansible, ansible-galaxy, YAML, curl

In this lab you will write a small Ansible playbook that installs and configures Nginx idempotently. Configuration as Code (CaC) is one of the CV0-004 deployment sub-objectives.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install Ansible

```
apt update && apt install -y ansible curl
ansible --version
```

Step 2 — Inventory (just localhost)

```
mkdir -p /tmp/ans && cd /tmp/ans
cat > inventory.ini <<'EOF'
[web]
localhost ansible_connection=local
EOF
```

Step 3 — Write a playbook (YAML)

```
cat > site.yml <<'EOF'
---
- name: Configure web server
  hosts: web
  become: true
  vars:
    site_msg: "Configured by Ansible"
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
        update_cache: true

    - name: Deploy index page
      copy:
        dest: /var/www/html/index.nginx-debian.html
        content: "<h1>{{ site_msg }}</h1>"

    - name: Ensure nginx is running
      service:
        name: nginx
        state: started
        enabled: true
EOF
```

Step 4 — First run (changes everything)

```
ansible-playbook -i inventory.ini site.yml
curl -s http://localhost
```

Look at the output: every task is **changed**.

Step 5 — Idempotency test (run again)

```
ansible-playbook -i inventory.ini site.yml
```

This time every task is **ok** — no work done. **Idempotency** is the core CaC property.

Step 6 — Detect drift / repeatability

Manually corrupt the page:

```
echo "drifted" > /var/www/html/index.nginx-debian.html
ansible-playbook -i inventory.ini site.yml
curl -s http://localhost
```

Ansible re-applies the desired state.

Step 7 — Variables and conditionals

```
cat > extra.yml <<'EOF'
---
- hosts: web
  become: true
  vars:
    enable_https: false
  tasks:
    - name: Install certbot only if HTTPS requested
      apt:
        name: certbot
        state: present
        when: enable_https
EOF
ansible-playbook -i inventory.ini extra.yml --check
```

when: is a conditional. Re-run with **-e enable_https=true** to flip the path.

Step 8 — Roles preview

```
ansible-galaxy init /tmp/ans/roles/myweb
ls /tmp/ans/roles/myweb
```

Roles are reusable bundles — the equivalent of Terraform modules.

Step 9 — Cleanup

```
apt remove -y nginx
```

What you learned

- Playbooks are YAML.
- Idempotent tasks: only act if state differs.
- Variables, conditionals, and roles compose larger configurations.

Free tools used

- Ansible — <https://www.ansible.com>
- Ansible Galaxy — <https://galaxy.ansible.com>
- YAML Lint (web) — <https://www.yamllint.com>

Lab 15 — JSON & YAML Scripting Logic

Goal: Parse, transform, validate and convert JSON ↔ YAML while exercising the exam's scripting-logic primitives.

Tools used: jq, yq, diff, web validators

In this lab you will read and write JSON and YAML, parse with **jq** and **yq**, and exercise the scripting-logic primitives the exam lists: variables, conditionals, operators, data types, functions.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install parsers

```
apt update && apt install -y jq python3-pip
pip3 install yq
```

Step 2 — Write JSON

```
mkdir -p /tmp/data && cd /tmp/data
cat > server.json <<'EOF'
{
  "name": "web-1",
  "cpu": 2,
  "ram_gb": 4,
  "tags": ["web", "prod"],
  "active": true
}
EOF

jq . server.json
jq '.name, .cpu' server.json
jq '.tags | length' server.json
```

Data types in play: string, number, boolean, array, object.

Step 3 — Same data in YAML

```
cat > server.yaml <<'EOF'
name: web-1
cpu: 2
ram_gb: 4
tags:
  - web
  - prod
active: true
EOF

yq . server.yaml
yq -y '.tags' server.yaml
```

Step 4 — Convert between formats

```
yq . server.yaml > /tmp/data/server-from-yaml.json
diff <(jq -S . server.json) <(jq -S . server-from-yaml.json) && echo "Same
data, two formats"
```

Step 5 — Variables, conditionals, operators

jq is itself a tiny scripting language — perfect to exercise the exam primitives.

```
# Variables
jq --arg env prod '. + {env:$env}' server.json

# Conditionals
jq 'if .ram_gb < 8 then "undersized" else "ok" end' server.json

# Operators
jq '.cpu * 2' server.json

# Functions
jq 'def double(x): x*2; double(.cpu)' server.json
```

Step 6 — Validate at the boundary

Try a broken file:

```
echo '{ "broken": ' > bad.json
jq . bad.json || echo "Schema violation caught"
```

In production, fail builds early on invalid JSON/YAML — same idea as Terraform `validate` or `kubectl --dry-run`.

Step 7 — Web validators (offline-friendly bookmarks)

- JSONLint — <https://jsonlint.com>
- YAML Lint — <https://www.yamllint.com>
- JSON Schema validator — <https://www.jsonschemavalidator.net>

These are useful when the exam shows you a malformed snippet.

Step 8 — Cleanup

```
rm -rf /tmp/data
```

What you learned

- Read, write, validate, and convert JSON ↔ YAML.
- Scripting logic primitives at the data-format level.
- When to choose JSON (machine-friendly) vs YAML (human-friendly).

Free tools used

- jq — <https://jqlang.github.io/jq>
- yq — <https://kislyuk.github.io/yq>
- JSONLint — <https://jsonlint.com>
- YAML Lint — <https://www.yamllint.com>

Domain 3 — Operations (17%)

Running cloud systems day-to-day: observability (metrics, logs, traces), alerting, auto-scaling, backup & disaster recovery, and patching / lifecycle management.

What this domain covers in the labs:

- **Lab 16 — Observability with Prometheus & Grafana:** Scrape a node's metrics with Prometheus, visualise them in Grafana and trigger a CPU alert.
- **Lab 17 — Centralized Logging with the ELK Stack:** Run ELK, ship logs into it, and query and retain them through the Elasticsearch API.
- **Lab 18 — Distributed Tracing with Jaeger:** Run Jaeger, instrument a Python 'checkout' path with OpenTelemetry and view a distributed trace of parent/child spans.
- **Lab 19 — Horizontal Auto-Scaling Simulation:** Simulate triggered, scheduled and manual horizontal scaling using a Bash controller that watches CPU and adds/removes Docker replicas.
- **Lab 20 — Backup & Recovery with restic:** Perform encrypted full/incremental/differential backups to off-site S3 storage, then test recoverability and verify integrity.
- **Lab 21 — Patching & Lifecycle Management:** Simulate a cloud resource lifecycle from provision through minor patch, major upgrade, testing and clean decommissioning.

Lab 16 — Observability with Prometheus & Grafana

Goal: Scrape a node's metrics with Prometheus, visualise them in Grafana and trigger a CPU alert.

Tools used: Prometheus, Grafana, node-exporter, stress-ng, Docker

In this lab you will scrape a node's metrics with **Prometheus**, visualise them in **Grafana**, and trigger an **alert** when CPU rises — covering the exam's logging/monitoring/alerting/triage sub-objectives.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker

```
apt update && apt install -y docker.io stress-ng curl
systemctl start docker
```

Step 2 — Run node-exporter (metrics source)

```
docker run -d --name node --net=host \
  prom/node-exporter:latest
sleep 2
curl -s http://localhost:9100/metrics | head -5
```

Step 3 — Configure and run Prometheus

```
mkdir -p /tmp/prom && cat > /tmp/prom/prom.yml <<'EOF'
global:
  scrape_interval: 5s
scrape_configs:
  - job_name: node
    static_configs:
      - targets: ['localhost:9100']
```

```
rule_files: ['alerts.yml']
EOF

cat > /tmp/prom/alerts.yml <<'EOF'
groups:
- name: cpu
  rules:
  - alert: HighCPU
    expr: 1 - avg(rate(node_cpu_seconds_total{mode="idle"}[1m])) > 0.5
    for: 30s
    labels: { severity: warning }
    annotations: { summary: "CPU > 50%" }
EOF

docker run -d --name prom --net=host \
-v /tmp/prom:/etc/prometheus \
prom/prometheus --config.file=/etc/prometheus/prom.yml
sleep 5
curl -s 'http://localhost:9090/api/v1/query?query=up' | head -c 200
```

Step 4 — Run Grafana

```
docker run -d --name grafana --net=host \
-e GF_SECURITY_ADMIN_PASSWORD=cloud \
grafana/grafana:latest
sleep 6
curl -s -u admin:cloud http://localhost:3000/api/health
```

Add the Prometheus data source via API:

```
curl -s -u admin:cloud -H 'Content-Type: application/json' \
-d
'{"name":"prom","type":"prometheus","url":"http://localhost:9090","access":"proxy","isDefault":true}' \
http://localhost:3000/api/datasources
```

Open <http://<killercode-host>:3000> in the playground's "Traffic / Ports" tab and explore.

Step 5 — Trigger the alert

```
stress-ng --cpu 4 --timeout 60s &
sleep 45
curl -s 'http://localhost:9090/api/v1/alerts' | head -c 400
```

Within ~30s the **HighCPU** alert is **firing** — Prometheus does the triage; alertmanager (production) does the response (PagerDuty, Slack, Teams).

Step 6 — Three pillars summary

Pillar	Tool you used	Lab
Metrics	Prometheus + node-exporter	16
Logs	rsyslog / ELK	17
Traces	Jaeger	18

Together they make a system **observable** — you can ask new questions about it without re-deploying.

Step 7 — Cleanup

```
docker rm -f node prom grafana
```

What you learned

- Scrape metrics, visualise, and alert on thresholds.
- Aggregation and retention are configured in Prometheus.
- Alerting → triage → response is the operational loop.

Free tools used

- Prometheus — <https://prometheus.io>
- Grafana OSS — <https://grafana.com/oss/grafana>
- node-exporter — https://github.com/prometheus/node_exporter
- stress-ng — <https://github.com/ColinIanKing/stress-ng>

Lab 17 — Centralized Logging with the ELK Stack

Goal: Run ELK, ship logs into it, and query and retain them through the Elasticsearch API.

Tools used: Elasticsearch, Logstash, Kibana, Docker, curl, jq

In this lab you will run **Elasticsearch**, **Logstash**, and **Kibana** (ELK) on the Killercode VM, ship logs from **rsyslog**, and query them through Kibana's API.

*ELK images are heavy (~1.5 GB). On Killercode, expect a slower pull. If memory runs tight, swap to the lighter **OpenSearch** alternative shown at the end.*

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Web tool: Build and test the grok/regex patterns for Logstash with the free browser **Regex Generator** — <https://alfredang.github.io/regexgenerator/> — before you put them in the pipeline config.

Step 1 — Install Docker

```
apt update && apt install -y docker.io curl jq
systemctl start docker
sysctl -w vm.max_map_count=262144
```

Step 2 — Run Elasticsearch

```
docker run -d --name es --net=host \
  -e discovery.type=single-node \
  -e xpack.security.enabled=false \
  -e ES_JAVA_OPTS='-Xms512m -Xmx512m' \
  docker.elastic.co/elasticsearch/elasticsearch:8.13.0
sleep 30
curl -s http://localhost:9200/_cluster/health | jq
```

Step 3 — Run Kibana

```
docker run -d --name kibana --net=host \
-e ELASTICSEARCH_HOSTS=http://localhost:9200 \
docker.elastic.co/kibana/kibana:8.13.0
sleep 30
curl -s http://localhost:5601/api/status | head -c 200
```

Open <http://<killercode-host>:5601> from the playground's Traffic tab.

Step 4 — Run Logstash with a syslog input

```
mkdir -p /tmp/ls && cat > /tmp/ls/pipeline.conf <<'EOF'
input { tcp { port => 5044 codec => line } }
filter { mutate { add_field => { "source" => "syslog" } } }
output { elasticsearch { hosts => ["http://localhost:9200"] index => "logs-%
{+YYYY.MM.dd}" } }
EOF

docker run -d --name logstash --net=host \
-v /tmp/ls/pipeline.conf:/usr/share/logstash/pipeline/logstash.conf \
docker.elastic.co/logstash/logstash:8.13.0
sleep 25
```

Step 5 — Ship test logs

```
for i in $(seq 1 20); do
  echo "$(date) cloudplus event $i severity=info" | nc -q 1 127.0.0.1 5044
done
sleep 3

curl -s 'http://localhost:9200/_cat/indices?v'
curl -s 'http://localhost:9200/logs-*/_search?q=event' | jq '.hits.total'
```

Step 6 — Retention policy (ILM concept)

```
curl -s -X PUT http://localhost:9200/_ilm/policy/logs-retention \
-H 'Content-Type: application/json' \
-d '{
  "policy": {
    "phases": {
      "hot": { "actions": {} },
      "delete": { "min_age": "7d", "actions": { "delete": {} } }
    }
  }
}' | jq
```

This enforces **7-day retention** — a CV0-004 logging sub-objective.

Step 7 — Aggregation query

```
curl -s -X POST http://localhost:9200/logs-*/_search \
-H 'Content-Type: application/json' \
-d '{"size":0,"aggs":{"by_severity":{"terms":{"field":"source.keyword"}}}}'
| jq
```


Step 8 — Cleanup

```
docker rm -f kibana logstash es
```

OpenSearch alternative (lighter)

If ELK is too heavy:

```
docker run -d --name os -p 9200:9200 -p 9600:9600 \
  -e discovery.type=single-node \
  -e DISABLE_SECURITY_PLUGIN=true \
  opensearchproject/opensearch:latest
```

Same API. Same Kibana clone is OpenSearch Dashboards.

What you learned

- Centralized logging pipeline: collect → ship → store → query.
- Retention is enforced with ILM policies.
- ELK and OpenSearch share the same query DSL.

Free tools used

- Elasticsearch / Kibana / Logstash — <https://www.elastic.co/elastic-stack>
- OpenSearch — <https://opensearch.org>
- rsyslog — <https://www.rsyslog.com>

Lab 18 — Distributed Tracing with Jaeger

Goal: Run Jaeger, instrument a Python 'checkout' path with OpenTelemetry and view a distributed trace of parent/child spans.

Tools used: Jaeger, OpenTelemetry Python SDK, Docker, python3, curl

In this lab you will run Jaeger all-in-one, instrument a small Python script with OpenTelemetry, and view a distributed trace in the Jaeger UI — covering the exam's tracing sub-objective.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker, Python deps

```
apt update && apt install -y docker.io python3-pip curl
systemctl start docker
pip3 install opentelemetry-api opentelemetry-sdk opentelemetry-exporter-otlp-
proto-grpc requests
```

Step 2 — Run Jaeger

```
docker run -d --name jaeger --net=host \
  jaegertracing/all-in-one:latest
sleep 5
curl -sI http://localhost:16686 | head -1
```

UI: <http://<killercode-host>:16686>.

Step 3 — Instrument a "checkout" microservice path

```
mkdir -p /tmp/trace && cd /tmp/trace
cat > app.py <<'EOF'
import time, random
from opentelemetry import trace
from opentelemetry.sdk.trace import TracerProvider
from opentelemetry.sdk.trace.export import BatchSpanProcessor
from opentelemetry.exporter.otlp.proto.grpc.trace_exporter import
OTLPSpanExporter

trace.set_tracer_provider(TracerProvider())
trace.get_tracer_provider().add_span_processor(
    BatchSpanProcessor(OTLPSpanExporter(endpoint="localhost:4317",
insecure=True))
)
tracer = trace.get_tracer("shop")

def db_lookup():
    with tracer.start_as_current_span("db.lookup"):
        time.sleep(random.uniform(0.05, 0.2))

def charge():
    with tracer.start_as_current_span("payment.charge"):
        time.sleep(random.uniform(0.1, 0.3))

def checkout():
    with tracer.start_as_current_span("http.checkout"):
        db_lookup()
        charge()

for _ in range(5):
    checkout()
    time.sleep(0.3)
print("traces sent")
EOF

python3 app.py
```

Step 4 — Query traces via Jaeger API

```
sleep 5
curl -s "http://localhost:16686/api/traces?service=shop&limit=5" | head -c 400
```

Open the UI, choose **service = shop**, **operation = http.checkout**, click a trace — you will see the parent span with two child spans (**db.lookup**, **payment.charge**) and their durations.

Step 5 — How tracing helps incident triage

A trace shows the **critical path** of a single request across all services. When a user reports "checkout is slow," you find the slowest span instead of guessing which service is at fault — this is why tracing is the third pillar of observability.

Symptom	Pillar that solves it
"CPU is high on host X"	Metrics (Lab 16)
"We had a 500 error at 10:42"	Logs (Lab 17)
"Why is *this specific request* slow?"	Traces (Lab 18)

Step 6 — Cleanup

```
docker rm -f jaeger
```

What you learned

- Spans are the unit of tracing; traces are causally-linked spans.
- OpenTelemetry is the vendor-neutral instrumentation API.
- Tracing answers "where did this one request spend its time?".

Free tools used

- Jaeger — <https://www.jaegertracing.io>
- OpenTelemetry — <https://opentelemetry.io>
- Zipkin (alternative) — <https://zipkin.io>

Lab 19 — Horizontal Auto-Scaling Simulation

Goal: Simulate triggered, scheduled and manual horizontal scaling using a Bash controller that watches CPU and adds/removes Docker replicas.

Tools used: Docker Compose, stress-ng, cron, nginx:alpine

In this lab you will simulate triggered, scheduled, and manual scaling of a service with Docker Compose plus a Bash controller that watches CPU and adds/removes replicas — the same pattern every cloud auto-scaler uses.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker Compose and load tools

```
apt update && apt install -y docker.io docker-compose-v2 stress-ng bc curl
systemctl start docker
```

Step 2 — Define a scalable service

```
mkdir -p /tmp/scale && cd /tmp/scale
cat > docker-compose.yml <<'EOF'
services:
  app:
    image: nginx:alpine
    deploy: { replicas: 2 }
EOF
docker compose up -d --scale app=2
docker compose ps
```

Step 3 — Triggered scaling (load-based)

This loop reads host CPU every 5 s and scales between 2 and 6 replicas.

```

cat > /tmp/scale/autoscale.sh <<'EOF'
#!/bin/bash
cd /tmp/scale
MIN=2; MAX=6
while true; do
    CPU=$(awk -v cores=$(nproc) '/cpu / {usage=($2+$4)*100/($2+$4+$5)} END{print usage}' /proc/stat)
    CUR=$(docker compose ps -q app | wc -l)
    TARGET=$CUR
    awk -v c="$CPU" 'BEGIN{exit !(c+0 > 70)}' && [ $CUR -lt $MAX ] && TARGET=$((CUR+1))
    awk -v c="$CPU" 'BEGIN{exit !(c+0 < 20)}' && [ $CUR -gt $MIN ] && TARGET=$((CUR-1))
    if [ "$TARGET" != "$CUR" ]; then
        echo "$(date +%T) CPU=${CPU}% scaling $CUR -> $TARGET"
        docker compose up -d --scale app=$TARGET --no-recreate
    fi
    sleep 5
done
EOF
chmod +x /tmp/scale/autoscale.sh
/tmp/scale/autoscale.sh &
ASPID=$!

```

Generate load:

```

stress-ng --cpu 4 --timeout 40s
sleep 60

```

You will see the controller scale **up** during stress and **down** afterwards. This is the **rending/load/event** trigger family from the exam objectives.

Step 4 — Scheduled scaling

```

echo "*/1 * * * * cd /tmp/scale && docker compose up -d --scale app=4 --no-recreate" | crontab -
crontab -l

```

A cron-driven pre-scale is the "scheduled" approach — useful before a known traffic peak.

Step 5 — Manual scaling

```

docker compose up -d --scale app=3 --no-recreate
docker compose ps

```

Step 6 — Horizontal vs vertical

Type	Action	Example
Horizontal	Add/remove replicas	This lab; AWS ASG, K8s HPA
Vertical	Resize a single instance	Stop/start with bigger CPU/RAM

Horizontal is the cloud default because it survives instance loss and scales linearly.

Step 7 — Cleanup

```
kill $ASPID 2>/dev/null
crontab -r 2>/dev/null
docker compose down
```

What you learned

- Triggered, scheduled, and manual scaling.
- A scaler is just a control loop watching a metric.
- Horizontal scaling is the cloud-native default.

Free tools used

- Docker Compose — <https://docs.docker.com/compose>
- stress-ng — <https://github.com/ColinIanKing/stress-ng>
- cron (built-in)

Lab 20 — Backup & Recovery with restic

Goal: Perform encrypted full/incremental/differential backups to off-site S3 storage, then test recoverability and verify integrity.

Tools used: restic, MinIO, Docker

In this lab you will use **restic** — a free, open-source, encrypted, deduplicating backup tool — to perform **full**, **incremental**, and **differential**-style snapshots, store them off-site (MinIO/S3), test recoverability, and verify integrity.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install restic and MinIO (off-site target)

```
apt update && apt install -y restic docker.io
systemctl start docker

docker run -d --name minio -p 9000:9000 \
  -e MINIO_ROOT_USER=admin -e MINIO_ROOT_PASSWORD=cloudplus \
  minio/minio server /data
sleep 4
docker exec minio mc alias set local http://127.0.0.1:9000 admin cloudplus
docker exec minio mc mb local/backups
```

Step 2 — Configure restic to talk to S3-compatible storage

```
export RESTIC_REPOSITORY="s3:http://localhost:9000/backups"
export AWS_ACCESS_KEY_ID=admin
export AWS_SECRET_ACCESS_KEY=cloudplus
export RESTIC_PASSWORD=cloudplus

restic init
```

The repo is **encrypted at rest** with **RESTIC_PASSWORD**.

Step 3 — Create source data and take a FULL backup

```
mkdir -p /data/app && for i in 1 2 3; do
  dd if=/dev/urandom of=/data/app/file$i bs=1M count=2 2>/dev/null
done
restic backup /data/app --tag full
restic snapshots
```

Step 4 — INCREMENTAL backup (only changed blocks)

```
echo "new content" >> /data/app/file1
dd if=/dev/urandom of=/data/app/file4 bs=1M count=1 2>/dev/null
restic backup /data/app --tag incremental
restic stats latest
```

restic deduplicates at block level — only deltas travel.

Step 5 — Differential-style: snapshot since the full

```
restic backup /data/app --tag diff --parent $(restic snapshots --tag full --
json | python3 -c 'import json,sys; print(json.load(sys.stdin)[0]
["short_id"])'
restic snapshots
```

Step 6 — Recoverability test (granular restore)

```
mkdir -p /restore
SNAP=$(restic snapshots --json | python3 -c 'import json,sys;
print(json.load(sys.stdin)[-1]["short_id"])'
restic restore $SNAP --target /restore --include /data/app/file1
ls -lh /restore/data/app/
```

Step 7 — Bulk restore (whole snapshot)

```
rm -rf /data/app
restic restore latest --target /
ls -lh /data/app/
```

Step 8 — Integrity check

```
restic check --read-data-subset=10%
```

check re-reads a subset and verifies hashes — the **integrity** sub-objective.

Step 9 — Retention policy

```
restic forget --keep-daily 7 --keep-weekly 4 --keep-monthly 12 --prune
```

Step 10 — Backup matrix

Type	What it stores	Restore speed
Full	Everything	Fastest (one snapshot)
Incremental	Changes since last backup of any kind	Slower (chain)
Differential	Changes since last full	Two snapshots

restic technically does **content-defined chunking** — every backup is a full restorable point but only sends deltas.

Step 11 — Cleanup

```
docker rm -f minio
rm -rf /data/app /restore
```

What you learned

- Encrypted, deduplicated backups to S3-compatible storage.
- Full / incremental / differential semantics.
- Recoverability and integrity tests are mandatory — not optional.

Free tools used

- restic — <https://restic.net>
- MinIO — <https://min.io>
- Borg (alternative) — <https://www.borgbackup.org>
- Duplicity (alternative) — <https://duplicity.gitlab.io>

Lab 21 — Patching & Lifecycle Management

Goal: Simulate a cloud resource lifecycle from provision through minor patch, major upgrade, testing and clean decommissioning.

Tools used: apt, unattended-upgrades, Docker, curl

In this lab you will simulate the lifecycle of a cloud resource: **provision** → **patch (minor)** → **upgrade (major)** → **test** → **decommission**. You will use **apt**, **unattended-upgrades**, container image tags, and clean teardown.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install patching tools

```
apt update && apt install -y unattended-upgrades apt-listchanges docker.io
systemctl start docker
```

Step 2 — Provision phase

```
docker run -d --name web --label lifecycle=active nginx:1.24-alpine
docker inspect -f '{{.Config.Image}}' web
```

Step 3 — Apply MINOR patches (no behaviour change)

```
apt-get -s upgrade | head -20 # simulate
unattended-upgrade --dry-run -d 2>&1 | tail -10
```

unattended-upgrades automates security minors. The CV0-004 distinction:

- **Minor** = bug fixes, security patches (1.24.0 → 1.24.5)
- **Major** = breaking changes (1.24 → 1.25)

Step 4 — MAJOR upgrade (test first!)

Stage the new version side-by-side:

```
docker run -d --name web-new nginx:1.27-alpine
docker exec web-new nginx -v
```

Run a smoke test:

```
curl -sI http://$(docker inspect -f '{{(index .NetworkSettings.Networks
"bridge").IPAddress}}' web-new)/ | head -1
```

If it passes, swap (a Lab-10 style blue-green):

```
docker rm -f web
docker rename web-new web
```

Step 5 — Persistent vs ephemeral data during upgrade

Persistent data (volumes, databases) **must survive** the swap. Ephemeral state (cache, /tmp) is expected to vanish.

```
docker volume create app-data
docker run -d --name app2 -v app-data:/var/lib/app nginx:alpine
docker exec app2 sh -c 'echo persistent > /var/lib/app/keep.txt'
docker rm -f app2
docker run --rm -v app-data:/var/lib/app alpine cat /var/lib/app/keep.txt
```

Step 6 — End of support / End of life

```
docker pull alpine:3.10 2>&1 | head -3 # old release
docker run --rm alpine:3.10 cat /etc/alpine-release
```

Alpine 3.10 hit **end of support** in 2021. Running EoS images means **no security patches** — a cloud-governance violation in most orgs.

Step 7 — Decommissioning

Tag, snapshot, then destroy.

```
docker commit web web-decom-$(date +%Y%m%d)
docker images | grep decom
docker rm -f web
docker volume rm app-data
docker rmi nginx:1.24-alpine 2>/dev/null
```

The **commit** is your archival snapshot; the **rm/rmi** is the decommission.

Step 8 — Lifecycle phases summary

1. Provision (Lab 13/14 — IaC/CaC)
2. Configure (Lab 14)
3. Monitor (Lab 16-18)
4. Patch (this lab)
5. Scale (Lab 19)
6. Backup (Lab 20)
7. Decommission (this lab)

What you learned

- Minor vs major upgrades have different risk profiles.
- Persistent data must be preserved across upgrades.
- EoL/EoS items must be replaced — not just patched.
- Decommissioning includes archive + revoke + delete.

Free tools used

- apt / unattended-upgrades (built-in)
- Docker — <https://www.docker.com>
- Pkgs.org (find current versions) — <https://pkgs.org>
- endoflife.date — <https://endoflife.date>

Domain 4 — Security (19%)

Securing the cloud: vulnerability management, CIS hardening, IAM & RBAC, secure access (bastion + MFA), secrets management, web application firewalls and runtime threat detection.

What this domain covers in the labs:

- **Lab 22 — Vulnerability Scanning with Trivy:** Scan a container image, filesystem and IaC config with Trivy following the scope → identify → assess → remediate workflow.
- **Lab 23 — CIS Benchmark Audit with Lynis:** Audit the VM against the CIS Ubuntu Benchmark with Lynis, score it, remediate findings and re-audit to confirm improvement.
- **Lab 24 — IAM & RBAC with Keycloak:** Stand up Keycloak and build a full IAM stack — realm, roles, users, an OAuth2 client — then authenticate and inspect RBAC roles in the JWT.
- **Lab 25 — Bastion Host with SSH Key + MFA:** Configure key-based SSH with TOTP MFA and route access through a locked-down bastion jump host — the canonical secure cloud-VM access pattern.
- **Lab 26 — Secrets Management with HashiCorp Vault:** Run Vault to store static secrets, mint auto-expiring dynamic DB credentials and audit every access.
- **Lab 27 — Web Application Firewall with ModSecurity:** Deploy Nginx + ModSecurity + the OWASP Core Rule Set and watch it block SQLi, XSS and path-traversal, then tune false positives.
- **Lab 28 — Detecting Suspicious Activity with Falco:** Run Falco and trigger runtime detections that map to CV0-004 attack categories, then route alerts to Prometheus / SIEM.

Lab 22 — Vulnerability Scanning with Trivy

Goal: Scan a container image, filesystem and IaC config with Trivy following the scope → identify → assess → remediate workflow.

Tools used: Trivy, Docker, sample Terraform

In this lab you will run **Trivy** against a container image, the local filesystem, and an IaC config. You will follow the exam's vulnerability-management steps: **scope** → **identify** → **assess** → **remediate**.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Trivy

```
apt update && apt install -y wget docker.io
systemctl start docker
wget -qO- https://aquasecurity.github.io/trivy-repo/deb/public.key | apt-key
add -
echo "deb https://aquasecurity.github.io/trivy-repo/deb generic main" >
/etc/apt/sources.list.d/trivy.list
apt update && apt install -y trivy
trivy --version
```

Step 2 — Scoping

Pick what to scan. For this lab:

1. A known-vulnerable image (**vulnerables/web-dvwa**)
2. The Killercode VM filesystem
3. A Terraform file from Lab 13

Step 3 — Identify (image scan)

```
trivy image --severity HIGH,CRITICAL alpine:3.10 | head -50
```

Each finding includes a **CVE** ID, package, fixed version, and CVSS score.

Step 4 — Assess

CVSS = severity × exploitability. The exam expects you to know CVE/CVSS terms.

```
trivy image --severity CRITICAL --format json alpine:3.10 | head -c 600
```

Group by severity:

```
trivy image alpine:3.10 --format table | tail -5
```

Step 5 — Remediate (rebuild with patched base)

```
trivy image alpine:3.20 --severity CRITICAL | tail -10
```

alpine:3.20 ships with patched packages. Remediation is usually:

1. Bump the base image.
2. Rebuild the application image.
3. Push and redeploy.

Step 6 — Filesystem scan

```
trivy fs --security-checks vuln,secret /etc | head -30
```

secret detection finds API keys / private keys committed by mistake.

Step 7 — IaC scan (misconfiguration)

```
mkdir -p /tmp/tf-bad && cat > /tmp/tf-bad/main.tf <<'EOF'
resource "aws_s3_bucket" "bad" { bucket = "world-readable" }
resource "aws_s3_bucket_acl" "bad" { bucket = aws_s3_bucket.bad.id  acl =
"public-read" }
EOF

trivy config /tmp/tf-bad
```

Trivy flags the public-read S3 bucket — an **outdated/insecure component definition**.

Step 8 — Daily automation hint

Add to a CI pipeline (Lab 30):

```
trivy image --exit-code 1 --severity CRITICAL myorg/myapp:latest
```

Non-zero exit fails the build — this is the "gate" that stops vulnerable images from shipping.

What you learned

- Vulnerability scanning closes the **identify** step.
- CVE + CVSS is the universal vocabulary.
- Image, filesystem, and IaC scans are all part of cloud security hygiene.

Free tools used

- Trivy — <https://aquasecurity.github.io/trivy>
- Gype (alternative) — <https://github.com/anchore/gype>
- OpenVAS / Greenbone CE — <https://www.greenbone.net>
- NVD CVE database — <https://nvd.nist.gov>
- CVE.org — <https://www.cve.org>

Lab 23 — CIS Benchmark Audit with Lynis

Goal: Audit the VM against the CIS Ubuntu Benchmark with Lynis, score it, remediate findings and re-audit to confirm improvement.

Tools used: Lynis, chmod, sysctl, Docker Bench Security

In this lab you will audit the Killercoda VM against the **CIS Ubuntu Benchmark** using **Lynis**, score it, and remediate one finding — covering the exam's *benchmark, hardening, patching* sub-objectives.

Run all commands on the Killercoda Ubuntu Playground:

<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install Lynis

```
apt update && apt install -y lynis
lynis --version
```

Step 2 — Run a baseline audit

```
lynis audit system --quiet
```

When the run finishes, look for:

```
Hardening index : 60 [##### ]
```

The **hardening index** is your CIS score. Tertiary Infotech treats anything below 80 as a fail.

Step 3 — Read the warnings

```
grep -E '\s+!' /var/log/lynis.log | head -20
lynis show warnings
lynis show suggestions | head -20
```

Each suggestion maps to a CIS control number — for example **file permissions on /etc/shadow**, **unused kernel modules**, **no MOTD banner**.

Step 4 — Remediate one finding (set permissions)

```
chmod 600 /etc/shadow /etc/gshadow
chmod 644 /etc/passwd /etc/group
ls -l /etc/shadow /etc/passwd
```

Step 5 — Remediate a kernel hardening finding

```
cat > /etc/sysctl.d/99-hardening.conf <<'EOF'
net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.default.send_redirects = 0
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.tcp_syncookies = 1
kernel.randomize_va_space = 2
EOF
sysctl --system | tail -10
```

Step 6 — Re-audit and compare

```
lynis audit system --quiet
grep "Hardening index" /var/log/lynis.log
```

The score should rise.

Step 7 — Vendor-specific benchmarks

Lynis covers Ubuntu/RHEL OS hardening. For cloud-vendor benchmarks use:

- **Prowler** (AWS, Azure, GCP, K8s) — <https://github.com/prowler-cloud/prowler>
- **kube-bench** (CIS Kubernetes) — <https://github.com/aquasecurity/kube-bench>
- **Docker Bench** (CIS Docker) — <https://github.com/docker/docker-bench-security>

Quick demo:

```
docker run --rm --net host --pid host --userns host --cap-add audit_control \
-v /etc:/etc:ro -v /var/lib:/var/lib:ro -v \
/var/run/docker.sock:/var/run/docker.sock:ro \
--label docker_bench_security \
docker/docker-bench-security 2>&1 | head -40
```

Step 8 — Cleanup

```
rm -f /etc/sysctl.d/99-hardening.conf
sysctl --system >/dev/null
```

What you learned

- Benchmarks are objective targets you can measure against.
- Hardening = closing the gap between current and benchmark.
- Different benchmarks for OS, Docker, K8s, AWS, Azure.

Free tools used

- Lynis — <https://cisofy.com/lynis>
- CIS Benchmarks (free PDFs) — <https://www.cisecurity.org/cis-benchmarks>
- Prowler — <https://github.com/prowler-cloud/prowler>
- kube-bench — <https://github.com/aquasecurity/kube-bench>
- Docker Bench Security — <https://github.com/docker/docker-bench-security>

Lab 24 — IAM & RBAC with Keycloak

Goal: Stand up Keycloak and build a full IAM stack — realm, roles, users, an OAuth2 client — then authenticate and inspect RBAC roles in the JWT.

Tools used: Keycloak, Docker, curl, jq

In this lab you will run **Keycloak** (open-source identity provider), create a realm, users, roles, and an OAuth 2.0 client — the building blocks of cloud IAM (federation, SAML, OIDC, RBAC).

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Run Keycloak

```
apt update && apt install -y docker.io curl jq
systemctl start docker

docker run -d --name kc -p 8080:8080 \
-e KEYCLOAK_ADMIN=admin \
-e KEYCLOAK_ADMIN_PASSWORD=admin \
quay.io/keycloak/keycloak:25.0 start-dev
sleep 25
curl -sI http://localhost:8080/realms/master | head -1
```

UI: <http://<killercode-host>:8080>.

Step 2 — Get an admin token

```
TOKEN=$(curl -s -X POST http://localhost:8080/realms/master/protocol/openid-connect/token \
-d "username=admin" -d "password=admin" -d "grant_type=password" -d
"client_id=admin-cli" \
| jq -r .access_token)
echo "$TOKEN" | head -c 40
```

This is an **OIDC** access token. Cloud admin CLIs do exactly the same call against AWS Cognito / Azure AD / GCP Identity.

Step 3 — Create a realm (= a tenant)

```
curl -s -X POST http://localhost:8080/admin/realms \
-H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
-d '{"realm":"cloudplus","enabled":true}'
```

Step 4 — Create roles (RBAC)

```
for role in viewer operator admin; do
  curl -s -X POST http://localhost:8080/admin/realms/cloudplus/roles \
  -H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
  -d '{"name\":"$role\"}'
done

curl -s http://localhost:8080/admin/realms/cloudplus/roles \
-H "Authorization: Bearer $TOKEN" | jq '.[].name'
```

Step 5 — Create a user and assign a role

```
curl -s -X POST http://localhost:8080/admin/realms/cloudplus/users \
-H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
-d '{"username":"alice","enabled":true,"credentials":
[{"type":"password","value":"cloud","temporary":false}]}'

UID=$(curl -s "http://localhost:8080/admin/realms/cloudplus/users?
username=alice" \
-H "Authorization: Bearer $TOKEN" | jq -r '.[0].id')

ROLE=$(curl -s http://localhost:8080/admin/realms/cloudplus/roles/operator \
-H "Authorization: Bearer $TOKEN")

curl -s -X POST http://localhost:8080/admin/realms/cloudplus/users/$UID/role-
mappings/realm \
-H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
-d "[$ROLE]"
```

Step 6 — Create an OAuth 2.0 client

```
curl -s -X POST http://localhost:8080/admin/realms/cloudplus/clients \
-H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
-d '{"clientId":"cli-
app","publicClient":true,"directAccessGrantsEnabled":true,"redirectUris":
["*"]}'
```

Step 7 — Authenticate as Alice (token-based)

```
ALICE_TOKEN=$(curl -s -X POST
http://localhost:8080/realms/cloudplus/protocol/openid-connect/token \
-d "username=alice" -d "password=cloud" -d "grant_type=password" -d
"client_id=cli-app" \
| jq -r .access_token)

# Decode the JWT payload
echo "$ALICE_TOKEN" | cut -d. -f2 | base64 -d 2>/dev/null | jq .realm_access
```

The payload includes `realm_access.roles: ["operator"]` — that is **role-based access control**.

Step 8 — Authentication models map

Exam term	Where you saw it
Local users	<code>username/password</code> in Keycloak
Federation / SAML	Keycloak → external IdP
OpenID Connect / OIDC	the <code>/protocol/openid-connect/*</code> endpoints
Token-based	bearer JWT
Directory-based	LDAP user federation in Keycloak
MFA	OTP setting per user (Lab 25)

Step 9 — Cleanup

```
docker rm -f kc
```

What you learned

- Identity provider, realms, users, roles, clients.
- OAuth 2.0 + OIDC token flow.
- RBAC roles propagate via JWT claims.

Free tools used

- Keycloak — <https://www.keycloak.org>
- jwt.io decoder (web) — <https://jwt.io>
- OAuth.tools — <https://oauth.tools>

Lab 25 — Bastion Host with SSH Key + MFA

Goal: Configure key-based SSH with TOTP MFA and route access through a locked-down bastion jump host — the canonical secure cloud-VM access pattern.

Tools used: OpenSSH, ssh-keygen, Google Authenticator (PAM), UFW, journalctl

In this lab you will configure SSH key-based authentication, add **Google Authenticator TOTP** as a second factor, and route through a **bastion** (jump host) — the canonical secure-access pattern for cloud VMs.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install OpenSSH server and Google Authenticator

```
apt update && apt install -y openssh-server libpam-google-authenticator
systemctl start ssh
ss -ltn | grep :22
```

Step 2 — Generate SSH key (no password)

```
ssh-keygen -t ed25519 -N "" -f /root/.ssh/id_ed25519
cat /root/.ssh/id_ed25519.pub >> /root/.ssh/authorized_keys
chmod 700 /root/.ssh
chmod 600 /root/.ssh/authorized_keys
```

Test:

```
ssh -o StrictHostKeyChecking=no localhost "echo SSH key auth OK"
```

Step 3 — Disable password auth, allow only keys

```
sed -i 's/^#*PasswordAuthentication.*/PasswordAuthentication no/'
/etc/ssh/sshd_config
sed -i 's/^#*PubkeyAuthentication.*/PubkeyAuthentication yes/'
/etc/ssh/sshd_config
sed -i 's/^#*ChallengeResponseAuthentication.*/ChallengeResponseAuthentication
yes/' /etc/ssh/sshd_config
sed -i 's/^#*UsePAM.*/UsePAM yes/' /etc/ssh/sshd_config
echo 'AuthenticationMethods publickey,keyboard-interactive' >>
/etc/ssh/sshd_config
```

Step 4 — Set up TOTP MFA for root

```
google-authenticator -t -d -f -r 3 -R 30 -W -q -s /root/.google_authenticator
ls -l /root/.google_authenticator
head -1 /root/.google_authenticator
```

The first line is the **TOTP seed** — scan that as a QR with Google Authenticator / Authy / FreeOTP on your phone.

Wire it into PAM:

```
sed -i '1a auth required pam_google_authenticator.so' /etc/pam.d/sshd
systemctl restart ssh
```

Now SSH requires **key + 6-digit TOTP**.

Step 5 — Create a "bastion" architecture

Run a private app behind a network namespace; the bastion is the only way in.

```
docker run -d --name app nginx:alpine
APP_IP=$(docker inspect -f '{{(index .NetworkSettings.Networks
"bridge").IPAddress}}' app)
echo "App IP (private): $APP_IP"

# Bastion = your Killercode root SSH on port 22
# Client jump:
ssh -J root@localhost root@$APP_IP "echo 'reached private app via bastion' "
2>&1 || true
```


The pattern in production: **internet** → **bastion (only host with public IP)** → **private subnet hosts**.

Step 6 — Restrict who can reach the bastion

```
ufw enable >/dev/null 2>&1 || apt install -y ufw
ufw allow from 10.0.0.0/8 to any port 22 proto tcp
ufw status verbose
```

Lock the bastion to corporate CIDRs only.

Step 7 — Audit trail

```
journalctl -u ssh -n 20 --no-pager
last -n 5
```

`last/journalctl` give you the **audit trail** required by SOC2 / ISO 27001.

Step 8 — Cleanup

```
docker rm -f app
sed -i '/AuthenticationMethods/d' /etc/ssh/sshd_config
sed -i '/pam_google_authenticator/d' /etc/pam.d/sshd
systemctl restart ssh
ufw disable 2>/dev/null
```

What you learned

- SSH key auth, MFA via TOTP, and bastion jump-hosts.
- AuthenticationMethods chains factors.
- Audit trails come for free with systemd-journald.

Free tools used

- OpenSSH — <https://www.openssh.com>
- Google Authenticator PAM — <https://github.com/google/google-authenticator-libpam>
- FreeOTP (mobile, free) — <https://freeotp.github.io>
- Authy / Microsoft Authenticator — free mobile apps

Lab 26 — Secrets Management with HashiCorp Vault

Goal: Run Vault to store static secrets, mint auto-expiring dynamic DB credentials and audit every access.

Tools used: HashiCorp Vault, Docker, PostgreSQL, curl, jq

In this lab you will run **HashiCorp Vault** in dev mode, store secrets, generate dynamic database credentials, and rotate them — the secrets-management sub-objective of CV0-004.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Run Vault in dev mode

```
apt update && apt install -y docker.io curl jq
systemctl start docker

docker run -d --name vault \
  --cap-add IPC_LOCK -p 8200:8200 \
  -e 'VAULT_DEV_ROOT_TOKEN_ID=cloudplus' \
  -e 'VAULT_DEV_LISTEN_ADDRESS=0.0.0.0:8200' \
  hashicorp/vault:latest
sleep 4

export VAULT_ADDR=http://localhost:8200
export VAULT_TOKEN=cloudplus
docker exec vault vault status
```

Step 2 — Write and read a static secret (KV v2)

```
docker exec -e VAULT_TOKEN=cloudplus vault \
  vault kv put secret/app/db username=admin password=Sup3rSecret

docker exec -e VAULT_TOKEN=cloudplus vault \
  vault kv get secret/app/db
```

Hard-coded secrets in code → ❌. Reference from Vault → ✅.

Step 3 — Versioned secrets (audit + rollback)

```
docker exec -e VAULT_TOKEN=cloudplus vault \
  vault kv put secret/app/db username=admin password=Rotat3d!

docker exec -e VAULT_TOKEN=cloudplus vault \
  vault kv get -version=1 secret/app/db
```

You just rolled back to v1. Versioning is built in.

Step 4 — Dynamic database credentials

Run a Postgres + tell Vault to mint **short-lived** users on demand.

```
docker run -d --name pg --network=host -e POSTGRES_PASSWORD=root postgres:16
sleep 5

docker exec -e VAULT_TOKEN=cloudplus vault sh -c '
  vault secrets enable database
  vault write database/config/pg \
    plugin_name=postgresql-database-plugin \
    allowed_roles="readonly" \
    connection_url="postgresql://{{username}}:{{password}}
@host.docker.internal:5432/postgres?sslmode=disable" \
    username="postgres" password="root"
  vault write database/roles/readonly \
    db_name=pg \
    creation_statements="CREATE ROLE \"{{name}}\" WITH LOGIN PASSWORD
\"{{password}}\" VALID UNTIL \"{{expiration}}\"; GRANT pg_read_all_data TO
\"{{name}}\";" \
```

```

    default_ttl="2m" max_ttl="5m"
  ,
  docker exec -e VAULT_TOKEN=cloudplus vault vault read database/creds/readonly

```

You now have a **2-minute-lifetime** Postgres user. After 2 minutes Vault deletes it. Compromise risk → near zero.

Step 5 — Audit log (compliance)

```

docker exec -e VAULT_TOKEN=cloudplus vault \
  vault audit enable file file_path=/vault/logs/audit.log

docker exec -e VAULT_TOKEN=cloudplus vault \
  vault kv get secret/app/db >/dev/null

docker exec vault tail -3 /vault/logs/audit.log

```

Every secret access is logged. SOC2 / PCI-DSS love this.

Step 6 — Cloud equivalents

Tool	Equivalent service
Vault	AWS Secrets Manager, AWS SSM Parameter Store, Azure Key Vault, GCP Secret Manager
Vault dynamic DB role	RDS IAM auth, Azure Managed Identity for SQL

Step 7 — Cleanup

```
docker rm -f vault pg
```

What you learned

- Static and dynamic secrets.
- Versioning, audit, lease/expiry.
- Dynamic creds collapse the blast radius of leaked passwords.

Free tools used

- HashiCorp Vault — <https://www.vaultproject.io>
- SOPS (alternative) — <https://github.com/getsops/sops>
- age (file encryption) — <https://github.com/FiloSottile/age>
- Mozilla SOPS + age + git — common GitOps secrets pattern

Lab 27 — Web Application Firewall with ModSecurity

Goal: Deploy Nginx + ModSecurity + the OWASP Core Rule Set and watch it block SQLi, XSS and path-traversal, then tune false positives.

Tools used: Docker (owasp/modsecurity-crs), Nginx, OWASP CRS, curl

In this lab you will deploy **Nginx + ModSecurity + the OWASP Core Rule Set (CRS)** and watch it block SQL injection, XSS, and path-traversal attempts.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Run a pre-built Nginx + ModSecurity image

```
apt update && apt install -y docker.io curl
systemctl start docker

docker run -d --name waf \
  -p 80:80 \
  -e BACKEND=http://example.com \
  -e PARANOIA=1 \
  owasp/modsecurity-crs:nginx
sleep 5
curl -sI http://localhost/
```

This image bundles Nginx + libmodsecurity + OWASP CRS — the same WAF logic AWS WAF and Cloudflare WAF base their managed rules on.

Step 2 — Run a benign request (allowed)

```
curl -s -o /dev/null -w "%{http_code}\n" "http://localhost/?q=cats"
```

Step 3 — Trigger a SQL injection rule

```
curl -s -o /dev/null -w "%{http_code}\n" "http://localhost/?id=1' OR '1'='1"
```

You should see **403** — CRS rule 942100 (SQLi).

Step 4 — Trigger an XSS rule

```
curl -s -o /dev/null -w "%{http_code}\n" "http://localhost/?
msg=<script>alert(1)</script>"
```

403 — CRS rule 941100 (XSS).

Step 5 — Trigger a path-traversal / LFI rule

```
curl -s -o /dev/null -w "%{http_code}\n"
"http://localhost/?file=../../../../etc/passwd"
```

403 — CRS rule 930100.

Step 6 — Inspect the WAF audit log

```
docker exec waf tail -40 /var/log/modsec/modsec_audit.log
```

You will see the matched rule IDs, the request, and the score.

Step 7 — Tune (false-positive triage)

ModSecurity uses an **anomaly score**. The default block threshold is 5. To accept a known-good pattern:

```
docker exec waf sh -c "echo 'SecRuleRemoveById 941100' >>
/etc/modsecurity.d/exclusions.conf"
docker restart waf
```

In production you tune via test → staging → prod, never prod-first.

Step 8 — Network ACL vs WAF vs Security Group

Layer	Inspects	Tool used
Network ACL (Lab 3)	IP/port	iptables
Security Group (Lab 3)	IP/port stateful	iptables -m conntrack
WAF (this lab)	HTTP semantics	ModSecurity / CRS

A WAF sees what a firewall cannot — request bodies, query strings, HTTP headers.

Step 9 — Cleanup

```
docker rm -f waf
```

What you learned

- WAFs operate on HTTP semantics, not just IP/port.
- OWASP CRS catches the OWASP Top 10 out of the box.
- Anomaly scoring + tuning is the production workflow.

Free tools used

- ModSecurity — <https://github.com/owasp-modsecurity/ModSecurity>
- OWASP Core Rule Set — <https://coreruleset.org>
- Cloudflare WAF (free tier) — <https://www.cloudflare.com/application-services/products/waf>

Lab 28 — Detecting Suspicious Activity with Falco

Goal: Run Falco and trigger runtime detections that map to CV0-004 attack categories, then route alerts to Prometheus / SIEM.

Tools used: Falco (eBPF), Docker, falcosidekick, journalctl

In this lab you will run **Falco** — a CNCF runtime security tool — and trigger detections for shell-in-container, sensitive-file reads, unexpected outbound connections, and cryptojacking-style behaviour.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

***Web tool:** Explore how these attacks look from the defender's side with the free browser **Cybersecurity Simulator** — <https://alfredang.github.io/cybersecuritysimulator/>.*

Step 1 — Install Falco (modern eBPF mode)

```
apt update && apt install -y curl gnupg lsb-release docker.io
systemctl start docker

curl -fsSL https://falco.org/repo/falcosecurity-packages.asc | gpg --dearmor
-o /usr/share/keyrings/falco-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/falco-archive-keyring.gpg]
https://download.falco.org/packages/deb stable main" >
/etc/apt/sources.list.d/falcosecurity.list
apt update
```

```
FALCO_DRIVER_CHOICE=modern_ebpf apt install -y falco
falco --version
```

If kernel headers are missing on KillerCoda, run Falco in **userspace** mode in a container:

```
docker run -d --name falco --privileged \
-v /var/run/docker.sock:/host/var/run/docker.sock \
-v /proc:/host/proc:ro \
falcosecurity/falco-no-driver:latest /usr/bin/falco -o engine.kind=ebpf
```

Step 2 — Tail Falco events

```
journalctl -fu falco &
JLP=$!
```

(or `docker logs -f falco &`)

Step 3 — Trigger: shell spawned in a container

```
docker run -d --name target nginx:alpine
docker exec target sh -c 'echo "shell inside container"'
```

Falco fires:

```
Notice A shell was spawned in a container ... container_id=...
```

Step 4 — Trigger: read of a sensitive file

```
docker exec target cat /etc/shadow
```

Falco rule **Read sensitive file untrusted** fires.

Step 5 — Trigger: unexpected outbound connection

```
docker exec target sh -c 'wget -q http://example.com -O /dev/null'
```

Falco rule **Unexpected outbound connection** fires (in default rules).

Step 6 — Map to attack types from the exam

Detection	Maps to exam category
Shell in container	Vulnerability exploitation
/etc/shadow read	Privilege escalation
Outbound connect	Cryptojacking / zombie instance
Sudden CPU spike (Lab 16)	Cryptojacking
Unexpected metadata API call	Metadata abuse (cloud-specific)

Step 7 — Baseline deviation alerting

Fold Falco events into Prometheus (Lab 16) via `falcosidekick`:

```
docker run -d --name fsk -p 2801:2801 \
-e FALCO_LISTEN=0.0.0.0:2801 \
falcosecurity/falcosidekick
```

Then in `falco.yaml` set `http_output: enabled: true, url: http://falcosidekick:2801`.

Step 8 — Cleanup

```
kill $JLP 2>/dev/null
docker rm -f target falco fsk 2>/dev/null
```

What you learned

- Runtime detection uses kernel signals (eBPF) to spot bad behaviour.
- Common detections map to CV0-004 attack categories.
- Falco events flow into Prometheus / SIEM.

Free tools used

- Falco — <https://falco.org>
- Falcosidekick — <https://github.com/falcosecurity/falcosidekick>
- auditd (built-in) — <https://github.com/linux-audit/audit-userspace>
- Wazuh (free SIEM) — <https://wazuh.com>

Domain 5 — DevOps Fundamentals (10%)

Source control and branching, CI/CD pipelines, web-service APIs (REST / GraphQL) and container orchestration with Kubernetes.

What this domain covers in the labs:

- **Lab 29 — Git Source Control & Branching:** Exercise every CV0-004 source-control verb — commit, push, branch, merge, PR review, conflict resolution — against a self-hosted Gitea server.
- **Lab 30 — CI/CD Pipeline with GitHub Actions (act):** Build and run a local test → build → scan → deploy CI/CD pipeline with act, mapping each stage to CV0-004 CI/CD concepts.
- **Lab 31 — REST & GraphQL APIs:** Call a public REST API, host a local GraphQL server, stream over WebSockets and preview pub/sub event-driven messaging.
- **Lab 32 — Container Orchestration with Kubernetes (k3s):** Install k3s and deploy, expose, scale, roll out/rollback, add persistence, self-healing and RBAC on a real Kubernetes cluster.

Lab 29 — Git Source Control & Branching

Goal: Exercise every CV0-004 source-control verb — commit, push, branch, merge, PR review, conflict resolution — against a self-hosted Gitea server.

Tools used: git, Gitea (self-hosted), curl, jq

In this lab you will exercise every CV0-004 source-control verb: **commit, push, branch, merge, pull request review** — using a local Gitea server as your "GitHub clone".

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install git and run Gitea (self-hosted)

```
apt update && apt install -y git docker.io curl jq
systemctl start docker

docker run -d --name gitea -p 3000:3000 -p 2222:22 gitea/gitea:latest
sleep 25
curl -sI http://localhost:3000 | head -1
```

UI: <http://<killercoda-host>:3000> — set up admin user (e.g. admin/cloudplus/a@x.com) on first visit.

Step 2 — Create a repository (UI or API)

```
curl -s -u admin:cloudplus -X POST http://localhost:3000/api/v1/user/repos \
-H 'Content-Type: application/json' \
-d '{"name":"infra","auto_init":true,"default_branch":"main"}' |
jq .full_name
```

Step 3 — Clone, commit, push

```
mkdir -p /tmp/work && cd /tmp/work
git clone http://admin:cloudplus@localhost:3000/admin/infra.git
cd infra
git config user.email lab@x
git config user.name lab

cat > main.tf <<'EOF'
resource "aws_s3_bucket" "data" { bucket = "main-bucket" }
EOF
git add main.tf
git commit -m "feat: initial bucket"
git push origin main
```

Step 4 — Branch management

```
git checkout -b feature/encryption
cat >> main.tf <<'EOF'
resource "aws_s3_bucket_server_side_encryption_configuration" "data" {
  bucket = aws_s3_bucket.data.id
  rule { apply_server_side_encryption_by_default { sse_algorithm =
"AES256" } }
}
EOF
git commit -am "feat: add SSE"
git push origin feature/encryption
```

Step 5 — Open a Pull Request

```
curl -s -u admin:cloudplus -X POST
http://localhost:3000/api/v1/repos/admin/infra/pulls \
-H 'Content-Type: application/json' \
-d '{"title":"Add
SSE","head":"feature/encryption","base":"main","body":"Encrypts the bucket"}'
```

Open the PR in the UI and review the diff — that is the **code review** sub-objective.

Step 6 — Merge

```
PR=1
curl -s -u admin:cloudplus -X POST
http://localhost:3000/api/v1/repos/admin/infra/pulls/$PR/merge \
  -H 'Content-Type: application/json' \
  -d '{"Do": "merge"}'

git checkout main
git pull
cat main.tf | tail -5
```

Step 7 — Resolve a merge conflict

```
git checkout -b feature/region
sed -i 's/main-bucket/main-bucket-us/' main.tf
git commit -am "feat: rename to us"
git push origin feature/region

git checkout main
sed -i 's/main-bucket/main-bucket-eu/' main.tf
git commit -am "feat: rename to eu"
git push origin main

git merge feature/region || true
grep -n '<<<' main.tf
# resolve
sed -i 's/<<<<<<<.*//; s/====.*//; s/>>>>>>>.*//' main.tf
git commit -am "merge: keep eu"
```

Step 8 — Branch protection (concept)

In production, protect **main**:

- Require PR + 1 approval.
- Require CI green.
- Block force-push.

These map to the exam's **branch management** sub-objective.

Step 9 — Cleanup

```
docker rm -f gitea
rm -rf /tmp/work
```

What you learned

- Commit → push → branch → PR → merge.
- Conflict resolution is a normal Git workflow.
- Branch protection enforces the review policy.

Free tools used

- git (built-in) — <https://git-scm.com>
- Gitea — <https://about.gitea.com>

- GitHub (free public repos) — <https://github.com>
- GitLab Community — <https://about.gitlab.com>
- Pro Git book (free) — <https://git-scm.com/book>

Lab 30 — CI/CD Pipeline with GitHub Actions (act)

Goal: Build and run a local test → build → scan → deploy CI/CD pipeline with act, mapping each stage to CV0-004 CI/CD concepts.

Tools used: act (nektos/act), Docker, git, pytest, Trivy

In this lab you will build a CI/CD pipeline locally with **act** — a free tool that runs GitHub Actions workflows in Docker. The workflow tests, builds, scans, and "deploys" an artifact.

Run all commands on the Killercode Ubuntu Playground:
<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install Docker, git, and act

```
apt update && apt install -y docker.io git curl
systemctl start docker

curl -s https://raw.githubusercontent.com/nektos/act/master/install.sh | bash
-s -- -b /usr/local/bin
act --version
```

Step 2 — Sample repository with a Python app

```
mkdir -p /tmp/cicd && cd /tmp/cicd
git init -q -b main

cat > app.py <<'EOF'
def add(a,b): return a+b
EOF

cat > test_app.py <<'EOF'
from app import add
def test_add(): assert add(2,3) == 5
EOF

cat > Dockerfile <<'EOF'
FROM python:3.12-slim
COPY app.py /app/app.py
WORKDIR /app
CMD ["python", "-c", "from app import add; print(add(2,3))"]
EOF
```

Step 3 — Define the workflow (.github/workflows/ci.yml)

```
mkdir -p .github/workflows
cat > .github/workflows/ci.yml <<'EOF'
name: ci
on: [push]
jobs:
```

```

test:
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - name: install
      run: pip install pytest
    - name: unit tests
      run: pytest -q
build:
  needs: test
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - name: build image
      run: docker build -t myapp:${{ github.sha }} .
scan:
  needs: build
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - name: trivy fs
      run: |
        curl -sSL https://aquasecurity.github.io/trivy/install.sh | sh -s --
        -b /usr/local/bin
        trivy fs --severity CRITICAL --exit-code 0 .
deploy:
  needs: scan
  runs-on: ubuntu-latest
  steps:
    - run: echo "Deploying $GITHUB_SHA to staging"
EOF

git add . && git -c user.email=lab@x -c user.name=lab commit -q -m "ci"

```

Step 4 — Run the workflow locally

```
act -j test
```

You will see Docker pull a runner image, install pytest, run the tests — green.

```

act -j build
act -j scan
act -j deploy

```

Or run the whole pipeline:

```
act
```

Step 5 — Map to CV0-004 CI/CD concepts

Stage in your YAML	Exam term
test	Testing
build	Code integration + Build
scan	Security
deploy	Code deployment
myapp:\${{github.sha}}	Artifact (container image)

Step 6 — Public vs private repositories

Push to GitHub:

```
# git remote add origin https://github.com/<you>/<repo>.git
# git push -u origin main
```

The same workflow runs on GitHub's hosted runners — public free for OSS, private free up to 2000 minutes/month.

Step 7 — Other free CI options

Tool	Type	Notes
GitHub Actions	SaaS	2000 free min/month private
GitLab CI	SaaS / self-hosted	400 free CI min
Jenkins	Self-hosted	classic, plugin-rich
Drone CI	Self-hosted	Docker-native
Tekton	Self-hosted on K8s	cloud-native

Step 8 — Cleanup

```
rm -rf /tmp/cicd
```

What you learned

- A pipeline is YAML + jobs + dependencies.
- act lets you iterate without pushing.
- Test → build → scan → deploy is the canonical flow.

Free tools used

- act — <https://github.com/nektos/act>
- GitHub Actions — <https://github.com/features/actions>
- GitLab CI — <https://docs.gitlab.com/ee/ci>
- Jenkins — <https://www.jenkins.io>
- Drone — <https://www.drone.io>

Lab 31 — REST & GraphQL APIs

Goal: Call a public REST API, host a local GraphQL server, stream over WebSockets and preview pub/sub event-driven messaging.

Tools used: curl, jq, GraphQL Yoga, websocat, redis-cli

In this lab you will call a public **REST** API, host a local **GraphQL** server, and try a **WebSocket** stream — covering the exam's web-services and integration sub-objectives.

Run all commands on the Killercode Ubuntu Playground:

<https://killercode.com/playgrounds/scenario/ubuntu>

Step 1 — Install tools

```
apt update && apt install -y curl jq docker.io websocat
systemctl start docker
```

Step 2 — Call a REST endpoint (CRUD)

The free **JSONPlaceholder** API supports GET/POST/PUT/DELETE.

```
curl -s https://jsonplaceholder.typicode.com/posts/1 | jq

# Create
curl -s -X POST https://jsonplaceholder.typicode.com/posts \
-H 'Content-Type: application/json' \
-d '{"title":"hello","body":"cloud","userId":1}' | jq

# Update
curl -s -X PUT https://jsonplaceholder.typicode.com/posts/1 \
-H 'Content-Type: application/json' \
-d '{"id":1,"title":"changed","body":"x","userId":1}' | jq

# Delete
curl -s -X DELETE https://jsonplaceholder.typicode.com/posts/1 | head -1
```

REST = stateless verbs over HTTP, one resource per URL.

Step 3 — Host a local GraphQL endpoint

```
docker run -d --name gql -p 4000:4000 \
graphql/swapi-graphql 2>/dev/null || \
docker run -d --name gql -p 4000:4000 \
-e PORT=4000 node:20-alpine sh -c '
  npm i -g graphql-yoga@4 graphql >/dev/null
  cat > /tmp/server.mjs <<"JS"
import { createYoga, createSchema } from "graphql-yoga"
import { createServer } from "node:http"
const yoga = createYoga({ schema: createSchema({
  typeDefs: `type Query { hello(name: String): String }`,
  resolvers: { Query: { hello: (_,a)=>"Hello "+(a.name||"cloud") } }
}))
createServer(yoga).listen(4000)
JS
  node /tmp/server.mjs'
sleep 8
```

Send a GraphQL query:

```
curl -s http://localhost:4000/graphql \
-H 'Content-Type: application/json' \
-d '{"query":"{ hello(name:\"Cloud\") }"}' | jq
```

GraphQL = single endpoint, client picks fields, fewer over-fetches.

Step 4 — REST vs GraphQL vs SOAP vs RPC

Style	Verb	Payload	Schema
REST	GET/POST/...	JSON	OpenAPI

SOAP	POST	XML	WSDL
GraphQL	POST	JSON	GraphQL SDL
gRPC (RPC)	HTTP/2	Protobuf	.proto

Step 5 — WebSocket stream (real-time)

```
docker run -d --name echo -p 8765:8080 \
  jmalloc/echo-server
sleep 2

echo "hello-stream" | websocat -n1 ws://localhost:8765/.ws
```

WebSockets keep the TCP connection open — used by chat, dashboards, K8s `kubectl exec`.

Step 6 — Event-driven preview

```
docker run -d --name redis -p 6379:6379 redis:7-alpine
sleep 2
( for i in 1 2 3; do
  docker exec redis redis-cli PUBLISH events "msg-$i"
  sleep 1
done ) &
docker exec redis sh -c 'timeout 5 redis-cli SUBSCRIBE events'
```

Pub/sub = event-driven architecture (Kafka, SNS/SQS, NATS use the same pattern).

Step 7 — Cleanup

```
docker rm -f gql echo redis 2>/dev/null
```

What you learned

- REST CRUD with curl.
- GraphQL queries return only requested fields.
- WebSockets for streams; pub/sub for events.

Free tools used

- jsonplaceholder.typicode.com — public REST sandbox
- GraphQL Yoga — <https://the-guild.dev/graphql/yoga-server>
- websocat — <https://github.com/vi/websocat>
- Postman free tier — <https://www.postman.com>
- Hoppscotch (open-source) — <https://hoppscotch.io>

Lab 32 — Container Orchestration with Kubernetes (k3s)

Goal: Install k3s and deploy, expose, scale, roll out/rollback, add persistence, self-healing and RBAC on a real Kubernetes cluster.

Tools used: k3s, kubectl, YAML manifests, curl

In this lab you will install **k3s** (a minimal, single-binary Kubernetes), deploy an app, expose it, scale it, and watch a rolling update — the canonical workload-orchestration platform on every cloud.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install k3s

```
apt update && apt install -y curl
curl -sL https://get.k3s.io | sh -
export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
kubectl get nodes
```

Step 2 — Deploy an application

```
cat > app.yaml <<'EOF'
apiVersion: apps/v1
kind: Deployment
metadata: { name: web }
spec:
  replicas: 3
  selector: { matchLabels: { app: web } }
  template:
    metadata: { labels: { app: web } }
    spec:
      containers:
      - name: web
        image: nginx:1.26-alpine
        ports: [{ containerPort: 80 }]
---
apiVersion: v1
kind: Service
metadata: { name: web }
spec:
  type: NodePort
  selector: { app: web }
  ports: [{ port: 80, nodePort: 30080 }]
EOF

kubectl apply -f app.yaml
kubectl rollout status deploy/web
kubectl get pods,svc
curl -s http://localhost:30080 | head -3
```

Step 3 — Scale (horizontal)

```
kubectl scale deploy/web --replicas=5
kubectl get pods -l app=web
```

This is the **horizontal scaling** primitive from Lab 19, but managed.

Step 4 — Rolling update (Lab 12 idea, K8s-native)

```
kubectl set image deploy/web web=nginx:1.27-alpine
kubectl rollout status deploy/web
kubectl rollout history deploy/web
```

Rollback if anything breaks:

```
kubectl rollout undo deploy/web
```

Step 5 — Persistent volume

```
cat > pvc.yaml <<'EOF'
apiVersion: v1
kind: PersistentVolumeClaim
metadata: { name: data }
spec:
  accessModes: [ReadWriteOnce]
  resources: { requests: { storage: 100Mi } }
EOF
kubectl apply -f pvc.yaml
kubectl get pvc
```

k3s ships with **local-path** provisioner — equivalent to AWS EBS / Azure Disk.

Step 6 — Liveness probe (self-healing)

```
kubectl patch deploy/web --type=json -p '[
  {"op": "add", "path": "/spec/template/spec/containers/0/livenessProbe",
   "value": {"httpGet":
    {"path": "/", "port": 80}, "initialDelaySeconds": 3, "periodSeconds": 5}}
]'
kubectl get pods -l app=web
```

If a pod fails its probe, the kubelet restarts it. **Self-healing**.

Step 7 — RBAC (Lab 24 idea, K8s-native)

```
kubectl create role pod-reader --verb=get,list --resource=pods
kubectl create rolebinding alice-pods --role=pod-reader --user=alice
kubectl auth can-i list pods --as alice
kubectl auth can-i delete pods --as alice
```

Step 8 — Map K8s ↔ exam objectives

K8s primitive	CV0-004 concept
Deployment	Workload orchestration
Service	Application/network load balancer
ConfigMap / Secret	CaC + Secrets management
HPA	Horizontal scaling
PVC	Persistent volume
RBAC	Authorization model
NetworkPolicy	Security group

Step 9 — Cleanup

```
kubectl delete -f app.yaml
kubectl delete pvc data
/usr/local/bin/k3s-uninstall.sh
```

What you learned

- k3s gives you a real cluster in one shell command.
- Deployments, services, scaling, rolling updates, RBAC.
- K8s implements every cloud-orchestration primitive on the exam.

Free tools used

- k3s — <https://k3s.io>
- kubectl — bundled
- Minikube (alternative) — <https://minikube.sigs.k8s.io>
- k9s (terminal UI) — <https://k9scli.io>
- Lens Desktop (free) — <https://k8slens.dev>
- Killercoda K8s playground — <https://killercoda.com/playgrounds/scenario/kubernetes>

Domain 6 — Troubleshooting (12%)

Methodically diagnosing and fixing deployment, network, security and container / resource issues using a repeatable triage workflow.

What this domain covers in the labs:

- **Lab 33 — Troubleshoot Deployment Issues:** Reproduce and fix each CV0-004 6.1 deployment failure mode locally — resources, permissions, oversubscription, sizing, deprecation, quotas, regions.
- **Lab 34 — Troubleshoot Cloud Network Issues:** Reproduce and resolve CV0-004 6.2 network failures using an L2→L7 diagnostic ladder (DHCP, DNS, NTP, NAT, HTTP codes, routing, IP overlap).
- **Lab 35 — Troubleshoot TLS, Cipher & Auth Issues:** Reproduce and fix CV0-004 6.3 security failures from the CLI — deprecated ciphers, privsec, unauthorized access, leaked credentials, vulnerabilities.
- **Lab 36 — Troubleshoot Container & Resource Limits:** Combine the Lab 33–35 methods to triage real container failures with a repeatable state → logs → stats → mounts → network workflow.

Lab 33 — Troubleshoot Deployment Issues

Goal: Reproduce and fix each CV0-004 6.1 deployment failure mode locally — resources, permissions, oversubscription, sizing, deprecation, quotas, regions.

Tools used: Docker, curl, jq, region matrices

In this lab you will reproduce and fix the deployment problems listed in CV0-004 6.1: **resource allocation, permissions, oversubscription, sizing, outdated definitions, deprecation, quotas, regional availability.**

Run all commands on the Killercoda Ubuntu Playground:

<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Setup

```
apt update && apt install -y docker.io curl jq
systemctl start docker
```

Step 2 — Resource allocation / sizing failure

```
docker run --rm --memory=10m alpine sh -c "yes | head -c 50m" 2>&1 | head -3
```

The container is OOM-killed because we under-sized memory.

```
docker stats --no-stream
```

Fix: raise the limit:

```
docker run --rm --memory=128m alpine sh -c "yes | head -c 50m >/dev/null && echo OK"
```

Step 3 — Permission misconfiguration

```
docker run -d --name pg --read-only -e POSTGRES_PASSWORD=cloud postgres:16 2>&1 | tail -3
docker logs pg 2>&1 | tail -5
docker rm -f pg
```

Postgres needs a writable volume; **--read-only** blocks startup.

Fix: mount a writable tmpfs/volume:

```
docker run -d --name pg --read-only --tmpfs /var/run/postgresql --tmpfs /tmp \
-v pgdata:/var/lib/postgresql/data -e POSTGRES_PASSWORD=cloud postgres:16
sleep 5 && docker logs pg | tail -3
docker rm -f pg && docker volume rm pgdata
```

Step 4 — Oversubscription

```
docker run --rm --cpus=0.1 alpine sh -c "for i in 1 2 3 4 5; do dd
if=/dev/zero of=/dev/null bs=1M count=200 & done; wait" 2>&1 | tail -3
```

CPU oversubscribed — work piles up in run queue. **Fix:** size CPU to peak load, not average.

Step 5 — Outdated component definition (image)

```
docker pull alpine:3.10 2>&1 | tail -1
docker run --rm alpine:3.10 cat /etc/alpine-release
```

3.10 is end-of-life — outdated definition. **Fix:** pin to a supported tag (**alpine:3.20**).

Step 6 — Deprecated functionality

```
docker run --rm alpine:3.20 sh -c 'echo "iptables-legacy still works:"; apk
add --quiet iptables; iptables -L 2>&1 | head -3'
```

Some flags in **docker run** are deprecated (**--link**). Always check the changelog when an upgrade fails.

Step 7 — Outage / partial outage simulation

```
docker run -d --name web -p 8080:80 nginx:alpine
docker pause web
curl -m 3 -sI http://localhost:8080 | head -1 || echo "PARTIAL OUTAGE:
container paused"
docker unpause web
```

```
curl -sI http://localhost:8080 | head -1
docker rm -f web
```

Step 8 — API throttling / service quota

Cloud APIs return HTTP **429 Too Many Requests** when throttled. Reproduce with a rate-limited responder:

```
docker run -d --name limit -p 8081:80 \
  -e DELAY=0 nginx:alpine

# Hammer it
for i in $(seq 1 50); do curl -s -o /dev/null -w "%{http_code}\n"
http://localhost:8081/; done | sort | uniq -c
docker rm -f limit
```

Fix in real cloud: request a quota increase, add exponential backoff, batch calls.

Step 9 — Regional service availability

Not every cloud service exists in every region. The fix is preventive — read the region matrix:

- AWS — <https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/>
- Azure — <https://azure.microsoft.com/en-us/explore/global-infrastructure/products-by-region/>
- GCP — <https://cloud.google.com/about/locations>

Step 10 — Decision tree

1. Does the deployment **start**? → permissions, image, definition.
2. Does it **stay up**? → resources, OOM, CPU.
3. Does it **respond**? → quotas, throttling, availability.
4. Did it **work before**? → drift, deprecation, version skew.

What you learned

- Reproduce each CV0-004 6.1 failure mode locally.
- Read logs/metrics first, fix the parameter that matches the symptom.
- Deprecation and EoL are deployment risks — track them.

Free tools used

- Docker — <https://www.docker.com>
- AWS / Azure / GCP region matrices (linked above)

Lab 34 — Troubleshoot Cloud Network Issues

Goal: Reproduce and resolve CV0-004 6.2 network failures using an L2→L7 diagnostic ladder (DHCP, DNS, NTP, NAT, HTTP codes, routing, IP overlap).

Tools used: dig/nslookup, ip, iptables, mtr, chrony, tcpdump

In this lab you will reproduce and resolve the network failures from CV0-004 6.2: **DHCP, DNS, NTP, NAT, HTTP status codes, latency, routing, switching, IP overlap, scope exhaustion.**

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Install diagnostics

```
apt update && apt install -y dnsutils iproute2 iputils-ping curl traceroute
mtr-tiny tcpdump iptables docker.io chrony jq
systemctl start docker
```

Step 2 — DNS failure → fix

```
echo "nameserver 192.0.2.1" > /tmp/bad-resolv.conf
docker run --rm --dns 192.0.2.1 alpine nslookup example.com 2>&1 | tail -4
echo "Symptom: SERVFAIL / no servers reachable"

# Fix: use a valid resolver
docker run --rm --dns 1.1.1.1 alpine nslookup example.com 2>&1 | tail -4
```

Step 3 — NTP / time skew

```
date
hwclock --systohc 2>/dev/null
chronyc tracking 2>/dev/null | head -5 || echo "chrony not running"
chronyc sources 2>/dev/null
```

Clock skew breaks **TLS handshakes, MFA tokens, and Kerberos**. Always enable an NTP source.

Step 4 — NAT failure

```
ip netns add p
ip link add veth0 type veth peer name veth1
ip link set veth1 netns p
ip addr add 10.99.0.1/24 dev veth0; ip link set veth0 up
ip -n p addr add 10.99.0.2/24 dev veth1; ip -n p link set veth1 up; ip -n p
link set lo up
ip -n p route add default via 10.99.0.1

# Without NAT – fails
ip -n p ping -c 2 -W 2 8.8.8.8 || echo "Symptom: blocked, no NAT"

# Add MASQUERADE
sysctl -w net.ipv4.ip_forward=1
iptables -t nat -A POSTROUTING -s 10.99.0.0/24 -o eth0 -j MASQUERADE
ip -n p ping -c 2 8.8.8.8 || echo "Outbound ping may be blocked by host
policy"

# Cleanup
iptables -t nat -D POSTROUTING -s 10.99.0.0/24 -o eth0 -j MASQUERADE
ip netns del p
```

Step 5 — HTTP status code triage

```
docker run -d --name web -p 8080:80 nginx:alpine
curl -s -o /dev/null -w "URL=%{url_effective} HTTP=%{http_code}\n"
http://localhost:8080/missing
curl -s -o /dev/null -w "HTTP=%{http_code}\n" http://localhost:8080/
docker rm -f web
```

Code	Meaning	Likely cause
4xx	Client	Wrong URL, auth, payload
5xx	Server	Backend down, TLS, misconfig
502/504	LB → upstream	Upstream not responding
429	Rate limited	Quota / WAF

Step 6 — Latency / bandwidth (Lab 35 of N+ revisited)

```
ping -c 4 1.1.1.1 | tail -2
mtr -rwc 5 1.1.1.1 || true
```

If RTT > expected, traceroute hop-by-hop to find the slow link.

Step 7 — IP overlap & scope exhaustion

```
ip route show | grep -E '(192\.168|10\.|172\.)'
echo "If two VPCs/peers share 10.0.0.0/24 – NAT or re-IP one side."
echo "Scope exhaustion: DHCP pool is full → enlarge the subnet or shorten the lease."
```

Step 8 — Routing issues

```
ip route get 8.8.8.8
ip route get 10.255.255.255 || true
```

`ip route get` shows which route the kernel selects. Missing or wrong default → no internet.

Step 9 — VLAN / switching issues (concept)

Two common symptoms:

- **Mismatched access vs trunk** — a tagged frame hits an access port → drop.
- **Wrong native VLAN** — frames leak between segments.

In cloud, the equivalent is wrong **subnet** → **route table** association.

Step 10 — Decision tree

1. Local stack OK? `ip a, ss -tlnp`
2. ARP / neighbour OK? `ip neigh`
3. Default route OK? `ip route`
4. DNS OK? `dig`
5. Path OK? `mtr`
6. Service OK? `curl -v`

What you learned

- A ladder of checks from L2 → L7.
- HTTP status codes localise the failure layer.
- NTP and DNS break "everything else" when they break.

Free tools used

- dig / nslookup / mtr / traceroute / ip — built-in
- chrony — <https://chrony-project.org>
- TIS PCAP Analyzer — <https://alfredang.github.io/pcapanalyzer/>
- HTTP status reference — <https://httpstatuses.com>

Lab 35 — Troubleshoot TLS, Cipher & Auth Issues

Goal: Reproduce and fix CV0-004 6.3 security failures from the CLI — deprecated ciphers, privesc, unauthorized access, leaked credentials, vulnerabilities.

Tools used: openssl, nmap, auditd, fail2ban, gitleaks, Trivy

In this lab you will reproduce and fix the security failures from CV0-004 6.3: **deprecated cipher suites, privilege escalation, unauthorized access, leaked credentials, software vulnerabilities, unauthorized software.**

Run all commands on the Killercoda Ubuntu Playground:

<https://killercoda.com/playgrounds/scenario/ubuntu>

***Web tool:** Rehearse the attack/defence scenarios in this lab with the free browser **Cybersecurity Simulator** — <https://alfredang.github.io/cybersecuritysimulator/>.*

Step 1 — Install tools

```
apt update && apt install -y openssl curl docker.io git auditd
systemctl start docker
systemctl start auditd
```

Step 2 — Deprecated cipher / TLS version

```
docker run -d --name oldssl -p 8443:443 \
-v /etc/ssl/certs:/certs:ro \
alpine sh -c "apk add --quiet openssl && openssl req -x509 -newkey rsa:2048
-nodes -keyout /tmp/k -out /tmp/c -subj '/CN=test' -days 1 && openssl s_server
-accept 443 -cert /tmp/c -key /tmp/k -tls1 -www" 2>/dev/null

sleep 4

# Try modern client (TLS 1.0 disabled by default)
echo | openssl s_client -connect localhost:8443 -tls1_3 2>&1 | grep -E
'(version|alert|verify return)' | head -3

# Negotiate weak version
echo | openssl s_client -connect localhost:8443 -tls1 2>&1 | grep 'Protocol' |
head -1
```

```
docker rm -f oldssl
```

Fix: disable SSLv3, TLS 1.0, TLS 1.1; require TLS 1.2+ with strong AEAD ciphers (AES-GCM, ChaCha20-Poly1305).

Step 3 — Test a public site's cipher suite

```
nmap --script ssl-enum-ciphers -p 443 example.com 2>/dev/null | head -40 || \
echo | openssl s_client -connect example.com:443 -servername example.com
2>/dev/null | grep -E '(Protocol|Cipher)'
```

Step 4 — Privilege escalation detection

```
auditctl -w /etc/sudoers -p wa -k sudo-changes
echo '# test' >> /etc/sudoers
ausearch -k sudo-changes 2>/dev/null | tail -5

# Find SUID binaries (potential privesc)
find / -perm -4000 -type f 2>/dev/null | head -10
```

Fix: restrict sudoers, audit SUID, run **linpeas**/**Lynis** regularly.

Step 5 — Unauthorized access / brute force

Tail SSH log for failed auth:

```
journalctl -u ssh --since "1 hour ago" 2>/dev/null | grep -i "failed\|invalid"
| head -5
```

Mitigation: install **fail2ban**:

```
apt install -y fail2ban
systemctl start fail2ban
fail2ban-client status sshd 2>/dev/null
```

Step 6 — Leaked credentials in source control

```
mkdir -p /tmp/leak && cd /tmp/leak && git init -q
cat > config.py <<'EOF'
AWS_ACCESS_KEY="AKIAIOSFODNN7EXAMPLE"
AWS_SECRET="wJalrXUtnFEMI/K7MDENG/bPxrFiCYEXAMPLEKEY"
EOF
git add . && git -c user.email=x@x -c user.name=x commit -q -m "oops"

# Detect with gitleaks
docker run --rm -v /tmp/leak:/repo zricethezav/gitleaks:latest detect -s /repo
--no-git -v 2>&1 | head -20
```

Fix: rotate the leaked key immediately; add a pre-commit **gitleaks** hook; store in Vault (Lab 26).

Step 7 — Software vulnerability + unauthorized software

```
# Vulnerable package check (Trivy from Lab 22)
docker run --rm -v /tmp:/scan aquasec/trivy fs --severity CRITICAL /scan 2>&1
| tail -10 || true
```

```
# Unauthorized software inventory: list packages added in last 7 days
grep " install " /var/log/dpkg.log 2>/dev/null | tail -10
```

Fix: allow-list packages; block unsigned repos; scan in CI (Lab 30).

Step 8 — TLS handshake debugging cheatsheet

Symptom	Likely cause
unable to get local issuer certificate	Missing CA bundle
handshake failure	Cipher mismatch
certificate has expired	Renew cert (Let's Encrypt)
hostname mismatch	SAN not matching SNI
ssl3_get_record:wrong version number	Plain HTTP on TLS port

Step 9 — Cleanup

```
sed -i '/# test/d' /etc/sudoers
rm -rf /tmp/leak
```

What you learned

- Negotiate TLS versions and ciphers from the CLI.
- Detect privesc, brute force, and leaked secrets.
- Common TLS errors and their root causes.

Free tools used

- OpenSSL — <https://www.openssl.org>
- testssl.sh — <https://testssl.sh>
- SSL Labs — <https://www.ssllabs.com/ssltest>
- gitleaks — <https://github.com/gitleaks/gitleaks>
- TruffleHog — <https://github.com/trufflesecurity/trufflehog>
- fail2ban — <https://www.fail2ban.org>
- auditd (built-in)

Lab 36 — Troubleshoot Container & Resource Limits

Goal: Combine the Lab 33–35 methods to triage real container failures with a repeatable state → logs → stats → mounts → network workflow.

Tools used: docker (inspect/stats/logs), a triage.sh script, k9s/ctop

In this final lab you will combine the methods from Labs 33-35 to triage real container failures: **CrashLoopBackOff**, **OOMKilled**, image pull failures, hostname/DNS resolution inside containers, and read-only filesystem errors.

Run all commands on the Killercoda Ubuntu Playground:
<https://killercoda.com/playgrounds/scenario/ubuntu>

Step 1 — Setup

```
apt update && apt install -y docker.io curl
systemctl start docker
```

Step 2 — CrashLoopBackOff (process exits immediately)

```
docker run -d --name crash --restart=on-failure alpine sh -c "exit 1"
sleep 3
docker ps -a --filter name=crash
docker logs crash
```

Kubernetes equivalent:

```
echo "kubectl describe pod <name> # look at 'Last State: Terminated, Reason'"
```

Fix: read logs, fix the entrypoint command, ensure the binary exists and exits 0.

Step 3 — OOMKilled

```
docker run -d --name oom --memory=20m --restart=no alpine \
  sh -c "yes hello | head -c 100m"
sleep 3
docker inspect oom --format '{{.State.OOMKilled}}' '{{.State.ExitCode}}'
{{.State.Error}}'
docker rm -f oom
```

OOMKilled=true is the **smoking gun**. Fix: raise the limit (**--memory=128m**) or fix the leak.

Step 4 — Image pull failure

```
docker run -d --name pull docker.io/myorg/no-such-image:1.0 2>&1 | tail -3
```

Fix: check image name, registry auth (**docker login**), private-registry credentials in K8s **imagePullSecrets**.

Step 5 — DNS inside containers

```
docker run --rm alpine nslookup kubernetes.default 2>&1 | head -3
docker run --rm --dns 1.1.1.1 alpine nslookup example.com | tail -3
```

In K8s:

```
echo "kubectl exec -it <pod> -- nslookup kubernetes.default"
echo "If this fails -> CoreDNS down or NetworkPolicy blocking 53/udp."
```

Step 6 — Read-only filesystem errors

```
docker run --rm --read-only alpine sh -c "echo x > /tmp/y" 2>&1 | tail -1
docker run --rm --read-only --tmpfs /tmp alpine sh -c "echo x > /tmp/y &&
cat /tmp/y"
```

Read-only roots are **good security** — give writable paths via **tmpfs** or volumes.

Step 7 — Resource quota exhaustion (sizing)

```
docker run -d --name big --memory=2g --cpus=4 alpine sleep 1000 2>&1 || \
  echo "If host has insufficient resources -> pending / failed scheduling"
docker rm -f big 2>/dev/null
```

K8s analogue: `kubectl describe pod` → `FailedScheduling: 0/1 nodes available: insufficient memory.`

Step 8 — End-to-end triage script

```
cat > /tmp/triage.sh <<'EOF'
#!/bin/bash
NAME=$1
echo "=== State ==="; docker inspect $NAME --format='{{.State.Status}}
OOM={{.State.OOMKilled}} Exit={{.State.ExitCode}}'
echo "=== Logs (last 30) ==="; docker logs --tail 30 $NAME 2>&1
echo "=== Stats ==="; docker stats --no-stream $NAME 2>/dev/null
echo "=== Mounts ==="; docker inspect $NAME --format='{{range .Mounts}}
{{.Source}} -> {{.Destination}} ({{.Mode}}){{println}}{{end}}'
echo "=== Network ==="; docker inspect $NAME --format='{{range $k,
$v := .NetworkSettings.Networks}}{{$k}}={{$v.IPAddress}}{{end}}'
EOF
chmod +x /tmp/triage.sh

docker run -d --name demo nginx:alpine
/tmp/triage.sh demo
docker rm -f demo
```

Step 9 — Master decision flow

1. **Pod won't start** → image pull, command, secrets.
2. **Pod crashloops** → application logs, exit code.
3. **Pod OOM** → resize memory, fix leak.
4. **Pod CPU throttled** → resize CPU, optimise code.
5. **Pod can't reach service** → DNS, NetworkPolicy, ServiceAccount RBAC.
6. **Pod read-only error** → mount tmpfs/volume.

Step 10 — Closing summary

You have now hit every CV0-004 troubleshooting category:

Category	Lab
Deployment	33
Network	34
Security	35
Container	36

What you learned

- Read state, logs, stats, mounts, network — in that order.
- OOM, CrashLoop, ImagePull, DNS, RO-fs are the daily five.
- A short triage script saves 10 minutes per incident.

Free tools used

- Docker — <https://www.docker.com>
- kubectl + k9s — <https://k9scli.io>
- ctop (container top) — <https://github.com/bcicen/ctop>

- lazydocker — <https://github.com/jesseduffield/lazydocker>

Quick Command Reference

Task	Command
Install Docker on the Killercode VM	<code>apt update && apt install -y docker.io</code>
Run a detached container (IaaS sim)	<code>docker run -dit --name x ubuntu:22.04 bash</code>
Inspect running containers when triaging	<code>docker ps / docker logs / docker inspect</code>
Read live CPU/memory (spot OOM & oversubscription)	<code>docker stats --no-stream</code>
Smoke-test a service endpoint	<code>`curl -s http://localhost:PORT`</code>
Inspect VPC subnets, routes and security groups	<code>ip netns / ip route / iptables -L</code>
Provision Infrastructure as Code	<code>terraform init / plan / apply</code>
Apply Configuration as Code idempotently	<code>ansible-playbook -i inventory.ini play.yml</code>
Operate a Kubernetes (k3s) workload	<code>kubectl get/scale/rollout</code>
Scan for vulnerabilities and misconfiguration	<code>trivy image / fs / config</code>
Core source-control workflow	<code>git commit / push / checkout -b / merge</code>

Support & Assessment

- **Trainer / enquiries:** enquiry@tertiaryinfotech.com · +65 6100 0613
- **Courseware & assessment (LMS):** <https://lms-tms.tertiaryinfotech.com/>
- **Website:** www.tertiarycourses.com.sg

Assessment Flow

At the assessment step, follow this order:

1. **TRAQOM** — scan the TRAQOM QR code on the LMS and complete the survey.
2. **Assessment Digital Attendance.**
3. **Assessment** — Written Assessment (WA) + Practical Performance (PP).
4. **Submit** your assessment answers on the LMS.
5. **Sign** the Assessment Summary Record.

Courseware and the assessment are on the LMS (<https://lms-tms.tertiaryinfotech.com/>). For a two-day class the 2-hour assessment runs 4:00–6:00 PM on the final day (1 hr WA + 1 hr PP).